Under consideration for publication in J. Fluid Mech.

Predictive reduced-order states for wall-pressure estimation of extreme vortex gust–airfoil interactions

Alberto Solera-Rico[1,2], Arnau Miró[3,4], Oriol Lehmkuhl[4] and Carlos Sanmiguel Vila[1,2]

[1] Aerial Platforms Department, Spanish National Institute for Aerospace Technology (INTA), San Martín de la Vega, 28330, Madrid, Spain

[2] Department of Aerospace Engineering, Universidad Carlos III de Madrid, Leganés, 28911, Madrid, Spain

[3] TUAREG Research Group, Universitat Politècnica de Catalunya (UPC), Carrer Colom 11, 08222 Terrassa, Spain

[4] Barcelona Supercomputing Centre (BSC), CASE Large-Scale Computational Fluid Dynamics, Plaça d'Eusebi Güell, 1-3, 08034 Barcelona, Spain

Corresponding author: Carlos Sanmiguel Vila, csanvil@inta.es

(Received xx; revised xx; accepted xx)

Extreme vortex gusts drive a post-stall airfoil through a transient reorganisation of its wake, and the release parameters alone do not determine the flow once the vortex has passed. We ask whether a reduced state of such an encounter can retain the wake dynamics, remain usable under a low-order forecast model, and be estimated from the sparse wall pressure a wing can carry. Direct numerical simulations of a NACA 0012 airfoil at $\alpha = 14^\circ$ and $\mathrm{Re} = 5000$, perturbed by Taylor vortices spanning gust ratio, core diameter and offset, are used to compare predictive, reconstruction-trained and proper-orthogonal-decomposition coefficient states at matched dimension. A controlled supervision study shows that these requirements are supplied by different ingredients. Under the predictive objective considered, a scalar lift target is necessary to prevent collapse of the latent representation, a near-body force-element target sharpens the peak loads, and a wake target is required to retain the wake enstrophy and circulation; the multi-step predictive objective contributes stability under rollout rather than instantaneous readability. Load information remains accessible at four coefficients, whereas wake-field reconstruction requires at least sixteen. Under a single direct multi-horizon forecaster trained identically on every state, the predictive coordinates are the most forecastable, autoregressive rollout degrades through error accumulation, and supplying the true gust parameters provides no benefit. A sequential ensemble filter sensing eight wall-pressure taps then tracks the lift through the encounter, with impact-phase $R^2 = 0.794$ (root-mean-square error 0.286 in lift units) on the in-distribution test set and $R^2 = 0.837$ at the held-out $|G| = 4$ boundary; the wake enstrophy is not recovered by the wall-sensor configurations considered, a limit set by the sensing geometry rather than by the state.

Key words: vortex shedding, low-dimensional models, machine learning

1. Introduction

Small uncrewed aerial vehicles are increasingly expected to operate in environments where the incoming flow is neither steady nor weakly disturbed. Urban canyons, mountainous terrain, coastal regions and ship or building wakes can expose these vehicles to gusts whose characteristic length scale is comparable to the wing chord and whose velocity is of the same order as the flight speed (Jones *et al.* 2022; Mohamed *et al.* 2023; Fukami & Taira 2023). In this regime, gusts cannot be treated as small perturbations of an otherwise steady aerodynamic state. They interact with separation, vortex shedding and wake recovery, producing large transient loads and rapid changes in the surrounding flow organisation. For small vehicles, this coupling directly affects navigability, control authority and flight safety.

This regime has motivated a growing body of work on extreme aerodynamics (Taira 2026). A canonical example is the interaction between a discrete vortex gust and a wing or airfoil, where the disturbance has sufficient strength to reorganise the near wake rather than merely modulate the instantaneous force (Fukami *et al.* 2025). Recent studies have shown that apparently complex gust encounters can exhibit unexpectedly low-dimensional structure when viewed in suitable coordinates. Lift-augmented autoencoders have revealed compact manifolds for extreme vortex-gust airfoil interactions (Fukami & Taira 2023), and the same reduced description has been used to support phase-amplitude analysis and data-driven transient lift attenuation (Fukami *et al.* 2024). Related work has also shown that large-scale vortical structures can display similarities across laminar and turbulent extreme-gust encounters (Odaka *et al.* 2026). These results suggest that the dynamical organisation of extreme aerodynamic flows may be simpler than the full flow-field dimension implies.

For practical deployment, however, compactness alone is not sufficient. A reduced description intended for gust mitigation must be usable as a plant state: it must be estimated from available sensors, advanced in time by a predictive model and queried by a controller or decision algorithm (Loiseau *et al.* 2018). This requirement places the problem closer to state estimation and model-based control than to flow-field compression alone. Surface pressure is especially important in this context, because it is one of the few measurements that can be obtained at high bandwidth on an aerodynamic body. Recent gust-modelling studies have therefore used sparse surface-pressure histories to infer unsteady aerodynamic loads and, more broadly, to bridge spatial pressure correlations and temporal flow memory (Zhong *et al.* 2023; Haughn *et al.* 2024; Chen *et al.* 2026), and Bayesian sequential estimation has begun to close the loop between sparse pressure and low-order aerodynamic states, with uncertainty carried alongside the estimate (Mousavi & Eldredge 2025; Eldredge & Mousavi 2025). Such approaches indicate a path towards deployable gust-response models, but they also sharpen the central question: what reduced state should be estimated from the wall in the first place?

Most current nonlinear reduced-order models for extreme aerodynamic flows are built around reconstruction-trained autoencoders. In this approach, a convolutional encoder maps the flow field to a low-dimensional latent vector, and a decoder reconstructs the original field. Aerodynamic observables, such as lift, can be added as auxiliary losses to orient the latent space towards physically relevant quantities (Fukami & Taira 2023, 2025). This strategy is powerful because it preserves a direct connection between the latent variables and the resolved flow. It also provides a natural route to visualisation and interpretation, since the decoder maps latent coordinates back to physical space. Similar ideas have been combined with variational autoencoders and sequence models to build predictive reduced-order models of chaotic flows (Solera-Rico *et al.* 2024; Fukagata & Fukami 2025).

Yet a reconstruction-trained latent space is not necessarily the best space for prediction. The decoder encourages the encoder to preserve information needed to reproduce the instantaneous field, but the time-advancing model requires coordinates that remain stable and informative under autoregressive rollout. These two requirements need not coincide. Recent comparisons in controlled wake flows indicate that convolutional-autoencoder-based predictors can be more prone to long-horizon divergence, whereas POD-based predictors, despite their linear and less compact representation, may retain higher median forecast accuracy and a narrower spread (Solera-Rico *et al.* 2025). This does not make POD preferable as a general representation of nonlinear separated flows. Rather, it exposes a tension: nonlinear encoders can compress more aggressively, but reconstruction compactness does not guarantee forecast robustness. The question is therefore whether one can construct nonlinear reduced states that retain the compactness of learned manifolds while improving their predictive usability.

This question motivates the use of a joint-embedding predictive architecture (JEPA) (LeCun 2022). In contrast with an autoencoder, a JEPA does not train the representation by reconstructing the input field. Instead, it learns an encoder and a predictor such that the present latent state predicts a future latent state. In computer vision, image- and video-based JEPAs have been introduced as non-generative self-supervised models that predict in representation space rather than pixel space (Assran *et al.* 2023, 2025). For the present fluid-mechanics problem, the relevance of JEPA is more specific: it provides a way to train the reduced coordinates in the same space in which the forward model will later operate. The architecture is therefore used here as a diagnostic tool to test whether predictive training can produce a nonlinear state that is more stable under rollout than a reconstruction-trained latent.

This choice creates a second question. Because the JEPA state is trained without a decoder, it is not obvious what physical information is retained by its coordinates. A low forecast error in latent space would be insufficient if the state did not carry the wake variables that matter for gust response. We therefore treat decoding not as part of the training objective, but as an a-posteriori diagnostic. After training the reduced state, we ask what can be recovered from it: aerodynamic observables, wake enstrophy, flow-field structures and, ultimately, the latent coordinates themselves from sparse wall pressure. This separation between predictive training and diagnostic decoding allows us to distinguish three properties that are often conflated: whether a state is compact, whether it is physically readable and whether it remains usable when advanced by its own dynamics.

The present work examines these questions using a canonical post-stall vortex-gust interaction. The gust parameters are not supplied to the predictive model; the encoder receives only the instantaneous flow state, and the predictor advances the latent state from its own history. We compare a predictive joint-embedding model with reconstruction-trained nonlinear encoders and a linear POD representation at matched latent dimension. The comparison is not designed to identify the method that reconstructs the vorticity field most accurately. Instead, it asks which representation supplies a reduced aerodynamic state that can be forecast, decoded and estimated in a form relevant to future gust-mitigation systems.

Here we ask whether a reduced representation of an extreme vortex-gust encounter can satisfy three requirements simultaneously: (i) it must carry the wake variables that distinguish post-impact states beyond the release parameters; (ii) it must remain usable when advanced by a low-order forecast model; and (iii) it must be estimable from the sparse wall pressure a vehicle carries. We compare predictive, reconstruction-trained and proper-orthogonal-decomposition coefficient states at matched dimension, on direct numerical simulations of a post-stall NACA 0012 airfoil perturbed by Taylor vortex gusts. Controlled ablations separate the roles of the lift, near-body and wake supervision targets; one direct multi-horizon forecaster, trained identically on every state, assesses forecastability; and a sequential estimator that senses only wall pressure tests observability. The requirements prove to be supplied by different ingredients: observable supervision organises the latent geometry and its readability, multi-step predictive training stabilises its forward use, and pressure histories recover the lift-carrying coordinates but not the complete wake state. The remainder of the paper is organised as follows. Section 2 describes the flow configuration, data and endpoints; §3 the reduced states, forecaster and estimators; §4 the results, in the order construction (§4.1), compression and forecastability (§4.2), wall observability (§4.3) and sequential estimation (§4.4); §5 discusses the mechanisms and their limits, and §6 concludes.

2. Flow configuration, data and endpoints

2.1. *Vortex gust-airfoil interaction*

We study a NACA 0012 airfoil at angle of attack $\alpha = 14^\circ$ and chord Reynolds number $\mathrm{Re} = 5000$, perturbed by a discrete vortex gust modelled as a spanwise-oriented Taylor vortex released upstream of the leading edge. The data analysed here are our own direct numerical simulations of this configuration, computed with the GPU-enabled spectral-element solver SOD2D (Gasparino *et al.* 2024) with no subgrid-scale model, so that all scales of the separated wake are resolved at this Reynolds number. The solver is run in the low-Mach regime in which the flow is effectively incompressible. The configuration and its physical characterisation follow Fukami, Smith & Taira (2025), which is the reference for the physics, not the source of the data. The vortex has the azimuthal velocity profile

$$u_\theta(r) = u_{\theta,\max}\,\frac{r}{R}\,\exp\left[\tfrac{1}{2}\left(1 - \frac{r^2}{R^2}\right)\right], \tag{2.1}$$

which peaks at the core radius $r = R$ (Taylor 1918), and is parametrised by three scalars: the signed gust ratio $G \equiv u_{\theta,\max}/u_\infty$, the core diameter $D \equiv 2R/c$, and the wall-normal offset Y of the vortex centre, the displacement normal to the chord (the laboratory coordinates are rotated by the angle of attack to define Y). The released-data case identifiers encode the solver's opposite sign convention, $s = -G$, so a case labelled `G+4.00` in the archive corresponds to $G = -4$ here. The profile and the (G, D, Y) sampling follow the configuration of Fukami *et al.* (2025). The training envelope spans $|G| \in \{0.5, 1.0, 1.5, 2.0, 3.0\}$ with $G = 0$ as the undisturbed reference, $D \in \{0.5, 1.0, 1.5\}$ chords, and $Y \in [-0.4, +0.4]$; the value $|G| = 4$ is held out as an extrapolation boundary, for reasons that are physical and that we return to below. Convective time t is measured in chords and set to zero when the vortex centre reaches the leading edge.

Without a gust, the $\alpha = 14^\circ$ wake at this Reynolds number is massively separated and sheds periodically; in dynamical terms the undisturbed flow is a limit cycle. A gust encounter is then a transient excursion from this limit cycle and back, and it is helpful to read it in stages (figure 1) (Smith *et al.* 2024; Fukami *et al.* 2025; Odaka *et al.* 2026). The vortex first induces a strong wall-normal vorticity flux at the leading edge, the airfoil experiences a sharp lift excursion as a leading-edge vortex (LEV) forms, the LEV rolls up and sheds together with a trailing-edge structure as the load reaches its extremum, and the wake then relaxes back toward the baseline shedding. The sign of G sets whether the first lift excursion is positive or negative, Y sets the side and strength of the impingement, D

sets how much of the encounter is dominated by the vortex core, and the impact timing sets the phase of the baseline cycle at which the disturbance arrives. The five observables defined in §2.2 are quantitative summaries of exactly this sequence.

A consequence of the source characterisation (Fukami *et al.* 2025) is central to how we read the extrapolation case. For $|G| \leq 3$ the encounter is primarily two-dimensional, so a mid-plane slice of the spanwise vorticity carries the relevant large-scale dynamics; for $|G| \geq 4$ the interaction becomes genuinely three-dimensional, with fine-scale instabilities developing during impact. We confirm this from the full three-dimensional vorticity fields: the fraction of the spanwise-vorticity (ω_z) enstrophy not captured by its spanwise mean, the content a mid-plane slice discards, is roughly constant across the training range ($|G| \leq 3$, post-impact wake median ≈ 0.20) and rises about threefold at $|G| = 4$ (≈ 0.56, over the four available cases). Because the encoder in this work consumes a single mid-plane vorticity field (§2.2), the held-out $|G| = 4$ case is not merely one parameter step beyond the training range; it is a regime in which the quantity the encoder observes no longer determines the true dynamics. We therefore treat $|G| = 4$ as a physical observability boundary rather than as a claim of parametric generalisation.

2.2. *Data, fields, and observables*

Each simulation case is one long run partitioned into encounters of 120 cached frames at $\Delta t = 0.05\, t/c$, one encounter per gust release. The simulations use a spectral-element discretisation of polynomial order $p = 4$, and the fields are cached on a structured grid of $192 \times 96 \times 32$ points spanning $x/c \in [-1.5, 4.5]$ and $y/c \in [-1.5, 1.5]$ with a spanwise extent of approximately one chord; this analysis subdomain matches the data-driven subdomain of the source characterisation (Fukami *et al.* 2025). The lift and drag coefficients are $C_L = F_L/(\frac{1}{2}\rho u_\infty^2 c)$ and $C_D = F_D/(\frac{1}{2}\rho u_\infty^2 c)$, obtained by integrating the pressure and viscous stresses over the airfoil surface. The remaining DNS solver-resolution details required for full reproducibility, the free-stream Mach number, the domain and spanwise extent, the element and solution-point counts, the near-wall spacing, the time step and CFL, the gust-release station, and the grid and time-step sensitivity, are listed as an author-fill checklist in table 1. The dataset comprises 102 cases partitioned into training, validation (the held-out last encounters of the training cases; archive name test_a), in-distribution test (test_b) and boundary test (test_c) sets; the archive names in parentheses label the corresponding records in the data release and appear only here. The in-distribution test set is stratified to span the gust strengths, the three core diameters, and the two source groups (figure 2); the boundary test set is the out-of-distribution $|G| = 4$ extrapolation set, sampled symmetrically in both signs of the gust (§2.2). These 102 cases are separated in preprocessing into 450 encounter windows of 120 frames each, one per gust release: 268 training, 100 validation, 42 in-distribution test and 40 boundary test encounters. The undisturbed reference case is retained in training. Behind these encounter counts are far fewer distinct cases: the 42 in-distribution test encounters come from only 10 held-out cases (6 interior and 4 boundary tier) and the 40 boundary test encounters from 8 cases, with most cases contributing four to six encounters that share a baseline shedding history. Encounters of the same case are therefore not independent, so encounter-level intervals overstate the effective sample size; the headline comparisons are reported additionally with a case-clustered bootstrap and a case-level paired test (Appendix A). The partition is fixed once and used identically by every model and baseline.

The $|G| = 4$ extrapolation boundary is sampled symmetrically by combining the periodic archive cases with four additionally released cases at the opposite sign of the gust. We verified the additional cases against the archive by a physical consistency check on the load rather than by trusting the file metadata: at matched core diameter and offset, the two

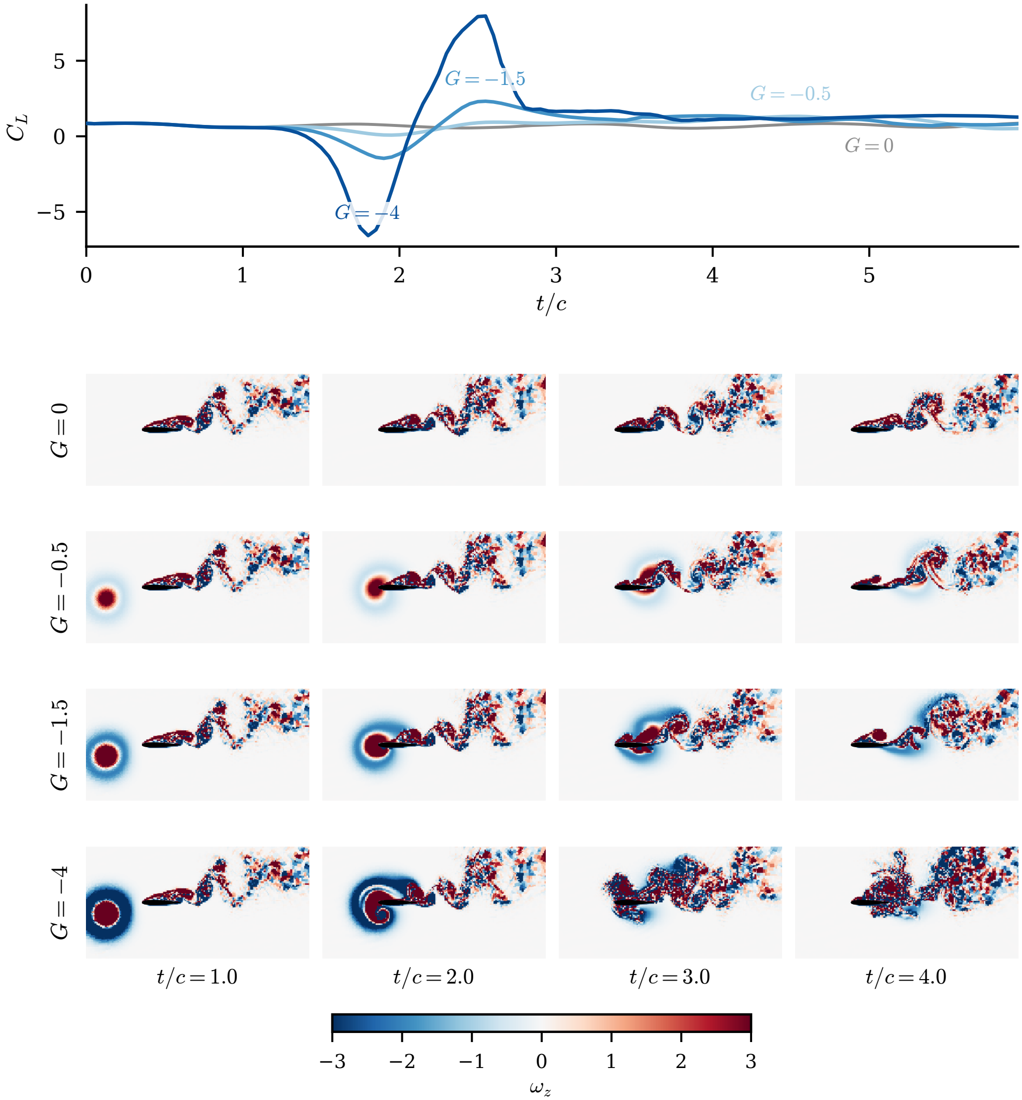


Figure 1. The vortex gust as a wake-reorganisation event, shown as a gust-strength sweep at fixed core diameter $D = 0.5$ and wall-normal offset $Y = -0.1$ (the first encounter of each case). (a) The lift coefficient C_L against convective time t/c for the four cases, with $t/c = 0$ the gust-release instant and impact near $t/c = 2$; the undisturbed baseline ($G = 0$) is grey, and the load excursion deepens and widens monotonically with the gust strength $|G|$. (b) The mid-plane spanwise vorticity ω_z for each case (rows, $G = 0, -0.5, -1.5, -4$) at four convective times (columns, $t/c = 1, 2, 3, 4$), in raw physical units with the colourbar clipped at ± 3 (the intense cores saturate; the wake structure away from them lies within the range), and the airfoil drawn as a filled polygon. The strong gust injects a vortex that reorganises the wake through impact, while the weak cases perturb the baseline shedding only slightly; $G = -4$ is the $|G| = 4$ extrapolation case, shown here only as a physical illustration. The load spike and the wake reorganisation are the surface and field signatures of the same event, and the release parameters alone do not fix where the wake sits afterwards, which is why the wake, not only C_L, is the endpoint a reduced state must carry.

Quantity	Value	Why a referee needs it
Free-stream Mach number (or incompressible-limit confirmation)	**[PENDING:]**	Confirms the low-Mach formulation is effectively incompressible at Re = 5000.
Computational domain and spanwise extent L_z/c	**[PENDING:]**	Bounds domain blockage and confirms the span resolves the three-dimensional wake.
Element and solution-point counts	**[PENDING:]**	Fixes the spatial resolution of the spectral-element discretisation.
Minimum wall-normal spacing $\Delta n_{\min}/c$ and wall units Δn^+	**[PENDING:]**	Shows the near-wall layer is resolved without a wall model.
Time step $\Delta t\, u_\infty/c$ and maximum CFL	**[PENDING:]**	Establishes the temporal resolution and stability of the integration.
Gust-release station x_0/c	**[PENDING:]**	Sets the upstream distance over which the Taylor vortex develops before impact.
Grid and time-step sensitivity check (baseline and a representative gust case)	**[PENDING:]**	Demonstrates that the reported statistics are converged.

Table 1. DNS solver-resolution details required for full reproducibility (§2.2). The simulations are performed with SOD2D without a subgrid-scale model; the solver-resolution metadata required to substantiate the DNS designation are listed here and will be supplied before submission. Values marked pending are owned by the simulation collaborators and are filled from `paper/dns_metadata.yaml` before submission.

signs of the $|G| = 4$ gust produce near mirror-image lift transients, one a positive-first excursion peaking near $+8.3$ and the other troughing near -7.6, as the sign of the gust ratio sets the sign of the first excursion (§2.1). This confirms that the two halves of the symmetric boundary set are physically consistent and that the sign-asymmetry analysis of the operating envelope (§4.4.2) is sound. Sampling the boundary at both signs is a genuine improvement over a one-sided $|G| = 4$ set: it lets the sign asymmetry be measured rather than assumed.

For each frame the cache stores the mid-plane spanwise vorticity $\omega_z(192, 96)$, the span-averaged wall pressure $p_{\mathrm{wall}}(192)$, and the lift and drag coefficients C_L, C_D. The vorticity is normalised by a fixed pipeline: a spatial mask removes the solid and one adjacent cell layer to suppress the leading-edge finite-difference artefact, a per-encounter high-percentile clip caps spikes, and the field is scaled by three times the training standard deviation ($\sigma_{\mathrm{train}} = 3.5396$) with no mean shift, preserving the vorticity antisymmetry. All representation and probe losses are computed in this normalised space; unnormalised values are used only for physical-units metrics and figures. The field-reconstruction similarity reported below uses the structural similarity index of Wang *et al.* (2004) with the standard constants $K_1 = 0.01$, $K_2 = 0.03$ and a dataset-dependent dynamic range $L = 8.487$, set to twice the global 99.9th percentile of the normalised target magnitude over the validation split so that the index is comparable across reruns. Appendix A repeats the primary wake probe under no amplitude clip and under a single training-pooled leak-free threshold (established on the previous-generation encoders; the pipeline is unchanged in this revision), and the family ordering is unchanged, so the per-encounter clip does not drive the result.

We evaluate forward closure against five physical observables that together span body-force and spatially distributed wake diagnostics. The lift and drag coefficients C_L and C_D are the body forces. The wake enstrophy and the positive and negative circulation are spatially distributed wake observables: they integrate (signed) vorticity over the wake region and are the most demanding to forecast, because they are sensitive to the redistribution of vorticity that the LEV roll-up and shedding produce. The choice is deliberate: the wake observables can change sharply while the scalar forces change little, so they are the part of the encounter most likely to separate a representation that has captured the gust-vortex physics from one that has only captured its force signature. On these grounds we fix two co-primary endpoints in advance: the wake enstrophy, which the reduced state must carry beyond the release parameters, and the lift coefficient C_L, which the sequential wall-pressure estimator of §3.3 tracks and in which the operating envelope is stated. The remaining three observables are reported as secondary and descriptive. This pre-registration is what the family-wide multiplicity correction in Appendix A is applied against.

The five observables are defined on each frame of the masked vorticity field ω_z described above. The lift and drag coefficients C_L, C_D are as defined earlier in this section. On the wake window $\Omega_w = \{(x, y) : x/c \in [0.5, 4],\ |y/c| \leq 1\}$ the wake enstrophy is

$$E_w(t) = \int_{\Omega_w} \omega_z^2 \,\mathrm{d}A, \tag{2.2}$$

and the signed wake circulations integrate the vorticity over its strongly rotational parts,

$$\Gamma^+(t) = \int_{\Omega_w,\ \omega_z > \omega_c} \omega_z \,\mathrm{d}A, \qquad \Gamma^-(t) = \int_{\Omega_w,\ \omega_z < -\omega_c} \omega_z \,\mathrm{d}A, \tag{2.3}$$

with threshold $\omega_c = 1$ in the cleaned vorticity units. The signed circulations at $\omega_c \in \{0.5, 1, 2\}$ are nearly collinear (pairwise Pearson well above 0.9 across held-out frames) and the predictive latent's circulation closure advantage over the reconstructive baseline is essentially unchanged across them, so the closure does not depend on this threshold. Every case uses the same fixed window and mask, and values are reported in the cache vorticity units integrated over chord-normalised area. The forecast horizon $H = 16$ is the sixteenth cached frame after impact, $t = 16\,\Delta t = 0.8\,c/u_\infty$.

One definitional point bounds how these observables are read. The wake and flow observables (the enstrophy and the two circulations) are computed on the same mid-plane slice the encoder consumes, so they omit the out-of-plane content quantified in §2.1 (roughly a fifth of the ω_z enstrophy in distribution), whereas C_L and C_D are true three-dimensional surface-integrated forces. The closure therefore compares mid-plane wake diagnostics against three-dimensional forces, and the mid-plane proxy is weakest at the $|G| = 4$ extrapolation boundary, where the out-of-plane fraction rises substantially.

3. Reduced states, forecasting and wall-pressure estimation

3.1. *Model and training objective*

The model takes no gust parameters: neither the encoder nor the predictor receives (G, D, Y). The predictive encoder E_θ maps the mid-plane vorticity field $\omega_z(192, 96)$ to a latent $\mathbf{z} \in \mathbb{R}^d$ at the headline dimension $d = 32$. It is a hybrid network: three convolutional downsampling stages produce a $24 \times 12 \times 256$ feature map, a six-layer vision transformer processes the resulting tokens, and the class-token output is projected to $\mathbf{z}$ through a single linear layer with a batch-normalisation boundary, which the characteristic-function regulariser below requires. The encoder is therefore a pure state map, and this is the property that lets the latent itself serve as the observable a controller would have access to, since at deployment the gust parameters are not known a priori (§5).

The predictor P_ϕ advances the pooled state in time, from its own history alone, so the model is unconditioned end to end. It is a vector sequence model: a six-layer autoregressive transformer with rotary temporal positions and a causal mask that maps a short history of 32-vectors to the next 32-vector, with no gust input. From the two most recent encoded

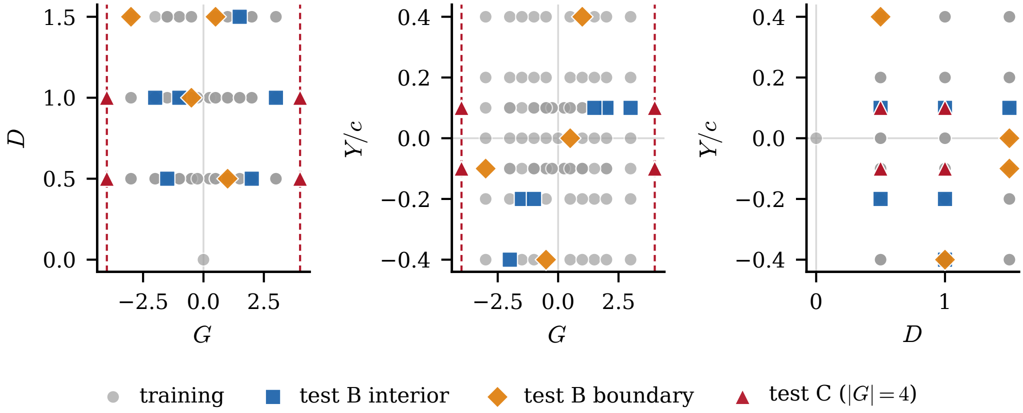


Figure 2. Parameter-space sampling of the 102 cases in the (G, D, Y) gust envelope, shown as three two-dimensional projections and coloured by split: training (268 encounters), the stratified in-distribution test set (42 encounters from 10 cases, split into interior and boundary tiers), and the boundary test extrapolation set (40 encounters from 8 cases). The dashed lines mark the $|G| = 4$ extrapolation boundary, now sampled at both signs of the gust so that the boundary set is symmetric rather than one-sided. Training mass concentrates near $Y/c = 0$, which is why the wall-normal offset is the least well resolved parameter (§4.1.4).

states it rolls eight steps open loop, feeding its own predictions back in, and the loss is the multi-step error between the rolled coefficients and the online encoder outputs at the matching frames, detached. The pipeline is therefore as plain as a linear or autoencoder reduced-order model: 32 numbers in, a dynamics model on the vector, 32 numbers out, with no spatial map anywhere in the loop. The state is the pooled vector at every stage: the anti-collapse regulariser, the observable heads, the probes, and the estimator all read the same 32-vector, and nothing in the model carries more. The estimator of §3.3 refits a predictor of this same class on the frozen pooled trajectories for its forecast step, so the training and estimation dynamics share a function class. Internal structure for both is given in Appendix A. Figure 4(a) shows the encoder and predictor, and panel (b) sets out the matched-predictor evaluation protocol used throughout; the contrast between the predictive training signal and the reconstructive one, in which the reconstructive autoencoder passes the field at time t through an encoder and a decoder and incurs a field-space error whereas the predictive model encodes the fields at t and $t+\Delta t$ with shared weights and incurs a latent-space error between the predicted and observed target embeddings, with no field-space loss, is drawn in Appendix A (figure 22). Table 2 summarises the two networks.

The predictive encoder and predictor are trained jointly on a multi-step open-loop rollout term, the predictor rolled eight steps from its own outputs with no teacher forcing against detached online targets, and an anti-collapse regulariser that prevents the encoder from collapsing to a constant. We use the characteristic-function regulariser of Balestriero & LeCun (2025); Maes *et al.* (2026), which matches the latent distribution to an isotropic Gaussian through a small number of random projections, with an automatic fall-back to a variance-covariance objective (Bardes *et al.* 2022) should the participation ratio of the latent indicate collapse. Its configuration is given in Appendix A.

Three auxiliary heads read aerodynamic observables off the encoder output during training: a lift head that maps $\mathbf{z}_t$ to $C_L(t)$, a wake head that maps $\mathbf{z}_t$ to an 80-dimensional signed wake spectrum, and a near-body head, constructed in §3.1.1 below, that maps $\mathbf{z}_t$ to an 80-dimensional lift-element observable of byte-matched capacity. All are trained with small weights jointly with the representation and none is part of the latent-prediction loss.

Table 2. Encoder and training-predictor configuration, counted from the released $d = 32$ checkpoint. The model is unconditional: no gust parameters $\mathbf{c} = (G, D, Y)$ enter the encoder or the predictor. Two small auxiliary heads read observables off the encoder output during training (a lift head and an 80-dimensional wake-spectrum head; §3.1). The estimation-stage forecast model of §3.3 is a predictor of the same class refitted on the frozen pooled trajectories.

	Encoder E_θ	Predictor P_ϕ
role	field → pooled state (unconditional)	pooled-state rollout (unconditional)
input	$\omega_z \in \mathbb{R}^{192\times 96}$	pooled states $\mathbf{z} \in \mathbb{R}^{32}$
backbone	3 conv. stages + 6-layer ViT	6-layer causal transformer (RoPE)
width / heads	256 / 8	384 / 16
image tokens / context	$24 \times 12 = 288$	2 frames, rolled 8 steps open loop
conditioning	none	none
state read-out	class token → linear → BatchNorm	next pooled state
parameters	6.7M	10.7M

The scalar-lift lineage is the observable augmentation of Fukami & Taira (2023, 2025); which head buys which property of the state is the subject of the conditioning study of §4.1.1. The wake target itself is a fixed descriptor chosen by design rather than tuned: it pairs a coarse signed spatial pooling of the wake region, which retains the layout of the alternating positive and negative vortices while discarding pixel-level detail, the spatial-pyramid construction of Lazebnik *et al.* (2006), with a radially-averaged power spectrum, the canonical isotropic scale descriptor of a turbulent field (Pope 2001) and an established compact image descriptor in machine learning (Durall *et al.* 2020), so it encodes both where the wake vortices sit and at which scales their energy resides. Pairing a localised spatial summary with a multiscale spectral one is the spatial-plus-scale representation formalised by the wavelet scattering transform (Bruna & Mallat 2013), which has been applied to turbulent flow fields precisely to recover geometry that the power spectrum alone cannot resolve (Skinner *et al.* 2025). We keep the descriptor compact, at 80 dimensions, so that it constrains the wake structure without over-parameterising the target on the available training encounters. The head-supervision axis is no longer a confound: the $2 \times 2 \times 2$ conditioning cube of §4.1.1, three seeds per cell with matched reconstructive anchors, isolates each head's contribution under pre-registered gates.

3.1.1. *The near-body lift-element head*

The third supervision head targets a physically constructed near-body observable built on the force-element decomposition of Chang (Proc. R. Soc. Lond. A 437, 1992), in the form popularised for vortex-dominated flows by Menon and Mittal (J. Fluid Mech. 918, 2021). An auxiliary potential ϕ_L is defined by the exterior boundary-value problem

$$\nabla^2 \phi_L = 0 \ \text{ outside the body}, \qquad -\mathbf{n}\cdot\nabla\phi_L = \mathbf{n}\cdot\mathbf{e}_L \ \text{ on the body}, \qquad \phi_L \to 0 \ \text{ far away}, \tag{3.1}$$

with $\mathbf{e}_L = (-\sin\alpha, \cos\alpha)$ the lift direction at $\alpha = 14°$. Equation (3.1) is discretised once on the 192×96 cache grid with a staircase immersed Neumann condition on the solid-adjacent cell faces and a homogeneous Dirichlet far field, and solved directly (figure 3); the interior Laplacian residual of the stored solution is 6.4×10^{-13} in the maximum norm. The lift-element density is then the pointwise field

$$e(\mathbf{x}, t) = \omega_z \left(-v\, \partial_x \phi_L + u\, \partial_y \phi_L\right), \tag{3.2}$$

evaluated from the raw mid-plane velocity and vorticity with the stored sign convention verified empirically (the cache stores $\partial u/\partial y - \partial v/\partial x$, so the standard ω_z is its negation).

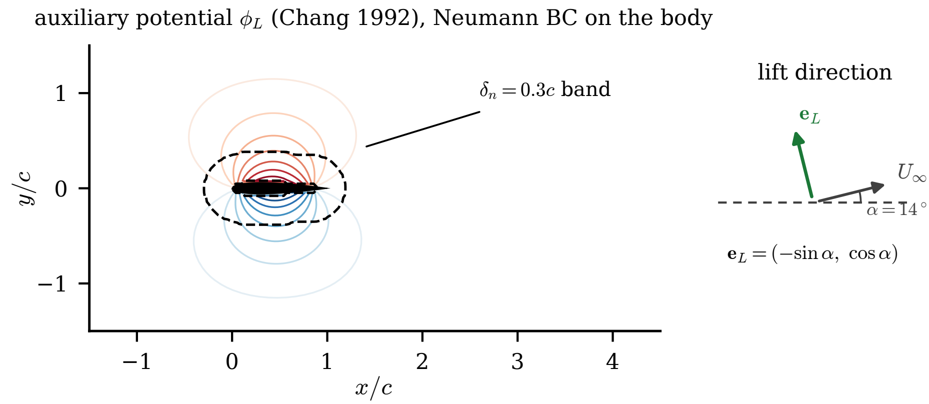
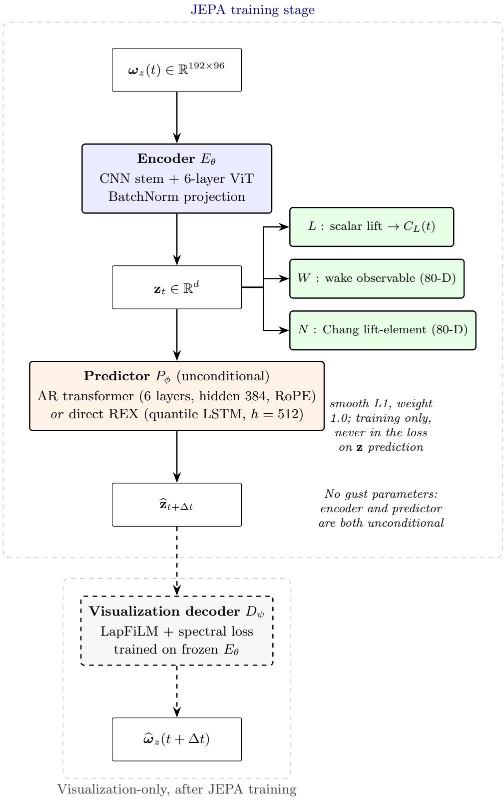


Figure 3. The Chang auxiliary potential ϕ_L of equation (3.1) on the cache grid (contours), the feathered $\delta_n = 0.3c$ supervision band (dashed) and the airfoil footprint; the inset shows the lift direction $\mathbf{e}_L$ against the inclined freestream at $\alpha = 14°$.

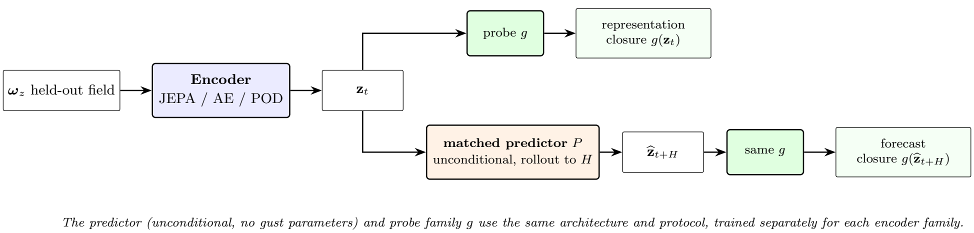
Because the cache window truncates the domain, the full-domain integral of (3.2) does not track the lift; the band-restricted field does, and the head therefore supervises e inside a feathered near-body band $w(\mathbf{x}) = \max\{0,\ 1 - \delta(\mathbf{x})/0.3c\}$, with δ the Euclidean distance to the body and its adjacent cells. The banded field $w\,e$, scaled by a fixed constant chosen so its 99th percentile magnitude is order one, is compressed to an 80-dimensional target by exactly the wake-observable construction: 64 sign-preserving patch energies (an 8×4 grid of $\log(1 + \mathrm{relu}(\pm x)^2)$ pooled energies) concatenated with a 16-bin Hann-windowed radial power spectrum. The two 80-dimensional targets are byte-matched in capacity, and the wake and near-body heads share one network class, so the conditioning comparison of §4.1.1 isolates the supervision signal, not the head architecture. A precompute quality gate requires the median lagged correlation between the band-integrated element and the stored lift over the gust-bearing training encounters (lag search $|\ell| \leq 25$ frames) to exceed 0.4; the achieved value is 0.74. The principled weighting is not interchangeable with a plain vorticity-magnitude proxy: the patch-block cosine between the two targets is 0.68 on the comparison sample.

3.2. *Matched-predictor closure protocol*

The comparison rests on a single principle: every baseline is the same pipeline, and only the encoder differs. Each encoder produces a latent trajectory $\mathbf{z}_{1:T}$ from the vorticity fields, and an identical probe family maps a latent to a physical observable for evaluation (figure 4(b)). Every latent is a pooled coefficient vector at the same dimension, so the same two forecast operators apply to all of them: a matched autoregressive transformer that reads the vector sequence directly, and a matched residual U-Net that tiles the vector to a grid and forecasts it convolutionally, both fitted with the same recipe and budget. They are not interchangeable in practice: the transformer is numerically stable on the pooled states of the controlled matrix but diverges when refitted on the reference latents, collapsing to a negative merit (Appendix A), while the U-Net is stable on all. We therefore report each latent under the operator that is stable on it, the transformer on the pooled states and the U-Net on the reference latents, rather than force one operator and either penalise the pooled states off-class or break on the references. The wake-supervised predictive state's own co-trained predictor, the as-built reduced-order model, is reported alongside, and the sequential estimator of §3.3 uses the same matched transformer on the wake-supervised predictive state's frozen trajectories. Every state is compared at the matched headline



(a)



(b)

Figure 4. The predictive model and how it is evaluated, drawn schematically (the training predictor is the vector transformer of table 2). (a) The encoder maps the mid-plane vorticity field to a latent $\mathbf{z}_t$, and the predictor, in either of its two variants (the vector transformer of table 2 or the direct multi-horizon forecaster of §3.2.1), advances the latent from its own history; no gust parameters enter the encoder or the predictor. The three supervision heads (L, W, N) read $\mathbf{z}_t$ during training with small weights and are never part of the latent-prediction loss. The visualisation decoder is trained only after the encoder is frozen and is used solely for the diagnostic decodes, never in the predictive loss. (b) Evaluation refits the same matched forecast operator and applies the same probe family g to every encoder family: a probe on the simulation-encoded latent measures representational closure, the headline, and the same probe on the rolled-out latent $\widehat{\mathbf{z}}_{t+H}$ measures forecast closure, reported as secondary.

dimension $d = 32$; the dimension plateau at $\{16, 32, 64\}$ and the minimum-dimension panel of §4.1.4 establish that this choice is a robustness statement rather than a tuned operating point. This isolation is the only fair way to attribute representation quality to the encoder rather than to incidental differences in the readout. The reconstructive references use their best-documented configuration, with ReLU activations and group normalisation rather than the strict published variant (tanh, no normalisation), which we found gives a worse parametric probe; the comparison is therefore against a tuned baseline, not a hobbled one. The observable heads are pinned to the same current-frame form for every family, so no family gains an accidental advantage from a differently-shaped supervision signal. The transformer-or-U-Net split above is retained as the legacy suited-operator protocol; the primary forecast-merit comparison of §4.2.3 uses one further operator, trained identically on every family's latents, so the family ordering cannot be an artefact of operator choice.

3.2.1. *The shared direct multi-horizon forecaster*

The forecast-merit comparison of §4.2.3 requires one operator trained identically on every family's latents; the operator used throughout is the direct multi-horizon forecaster, a quantile forecaster whose design transfers the deployable ingredients of the TiRex forecasting line (Auer *et al.* 2025; Podest *et al.* 2026) without its foundation-model machinery. Each context window of L consecutive states (L drawn uniformly from 16 to 30 during training, fixed at $L = 25$ at evaluation) is normalised per window, $\tilde{\mathbf{z}} = \operatorname{arcsinh}((\mathbf{z} - \mu)/\sigma)$ with μ, σ the window mean and standard deviation (floored at 10^{-3}), passed through a two-layer LSTM of width 512, and mapped by a single feed-forward head to the full forecast tensor of 40 horizon steps, d coordinates and nine quantiles $q \in \{0.1, \ldots, 0.9\}$ in one shot; the inverse transform restores physical scale. There is no autoregressive feedback, so rollout error cannot compound by construction. Training minimises the pinball loss

$$\mathcal{L} = \frac{1}{|Q|} \sum_{q \in Q} \left\langle \rho_q(z - \hat{z}_q) \right\rangle, \qquad \rho_q(u) = \max(qu,\ (q-1)u)\,, \tag{3.3}$$

with AdamW (learning rate 10^{-3}, weight decay 10^{-4}), cosine decay, batch 64, gradient clipping at unit norm, and 6000 iterations. The backbone and width were selected once on a held-out 15 % case split of the training pool over $\{\text{LSTM}, \text{sLSTM}\} \times \{256, 512\} \times \{3, 9\}$ quantiles by validation decoded-lift skill; a pure-PyTorch sLSTM cell with exponential gating and the log-domain stabiliser of Beck *et al.* (2024) ties the LSTM within seed noise and the selected operator is the LSTM at width 512 with nine quantiles. The median forecast is the point prediction everywhere; the predicted inter-quantile band supplies the state-dependent process noise of §3.3.3. The adopted, refuted and rejected ingredients of that source line, including the oracle-covariate negative of §4.2.3, are catalogued in the supplementary material.

The headline figure of merit is representational closure: a fixed linear probe is applied to the latent encoded directly from a held-out field, and we ask whether it recovers each of the five observables, the wake in particular. This isolates what the encoder carries, with no rollout. We report it as the held-out coefficient of determination R^2 on the in-distribution test and boundary test sets, with the per-observable mean absolute error alongside, and bootstrap 95% confidence intervals over encounters. As a secondary, confirmatory measure we also report forecast closure: the predictor is rolled recursively from the pre-impact context to $H = 16$, and the same probe maps the predicted latent to each observable. A small forecast error means the encoder produces a latent whose dynamics, propagated by the predictor, retain the information the probe needs; a large error means either that the encoder discarded that information or that the predictor lost it during rollout, a distinction the drift diagnostic below resolves. The representational closure is in table 6, which reports the readability and the matched-predictor forecast merit side by side. We do not headline a forecast-R^2-versus-horizon curve; the rollout is reported to show that the latent stays usable, not as a validated forecast.

3.3. *Estimating the state from the wall*

At deployment the vorticity field is unavailable and the wall pressure is not, by itself, enough to fix the state: a single pressure frame recovers little of the wake through any of the latents (§4.3). The reason a single frame is not enough, and a window is, is a statement about the dynamics rather than about the number of sensors. The encounter lies on an attractor of low effective dimension, a limit cycle with a gust excursion and a relaxation, into which the encoded latent concentrates its variance (§4.1.4). By the delay-embedding theorem (Takens 1981; Sauer *et al.* 1991), a generic observation of such a system, read over a window of delays, reconstructs its state up to a diffeomorphism, and with K simultaneous taps over m delays the reconstruction is generic once the product mK exceeds twice the attractor dimension. Because that dimension is small, a handful of taps over a modest window suffices, and the delay window can substitute for sensor count, a trade quantified in §4.3. A coordinate the instantaneous pressure misses, the wake circulation, is still reconstructed by the window provided it is dynamically coupled to what the pressure sees, which for the coupled vortex dynamics it is. The estimator must therefore be sequential, accumulating pressure innovations through learned dynamics, which is the setting of data assimilation.

Every estimator in this paper solves the same discrete state-estimation problem on the frozen coefficient state. Writing $\mathbf{z}_t \in \mathbb{R}^d$ for the encoded state at frame t (time base $\Delta t = 0.05\, c/u_\infty$, the cache frame spacing) and $\mathbf{p}_t \in \mathbb{R}^K$ for the wall pressure at the K selected taps, the pair

$$\mathbf{z}^f_{t+1} = F\left(\mathbf{z}^a_t\right) + \boldsymbol{\eta}_t, \qquad \boldsymbol{\eta}_t \sim \mathcal{N}(0, Q_t), \tag{3.4}$$

$$\mathbf{y}_t = h(\mathbf{z}_t) + \boldsymbol{\epsilon}_t, \qquad \boldsymbol{\epsilon}_t \sim \mathcal{N}(0, R), \tag{3.5}$$

defines each member of the suite once its four ingredients (F, h, Q_t, R) are named; the remainder of this subsection does exactly that, and table 4 collects every parameter. The frozen probes that read the lift and the wake enstrophy off the state can never enter the innovation: this leakage guarantee is enforced by construction and by a unit test, so the reported analysis closure is a genuine estimate, not a quantity the filter was given (figure 5). Two evaluation frames are used and always stated. The envelope frame assimilates over $[t_{\mathrm{imp}} - 24,\, t_{\mathrm{imp}} + 48]$ frames with a field-free initialisation at the window start; the ladder frame runs the full encounter with the ensemble initialised from the encoded observation at $t_0 = 9$ (the delay-embedding warm-up) and scores on the same impact window. All estimator fits, without exception, use the training split only.

3.3.1. *Base stochastic ensemble filter*

The frozen production filter is a stochastic ensemble Kalman filter (Evensen 2003) with $N = 64$ members. The forecast model F is the matched autoregressive transformer of §3.1, refitted on the frozen pooled training trajectories; each member carries its own buffer of past analysis states (at most 32 frames) and is advanced one step per frame. The process noise is state-independent, $Q_t \equiv Q$, estimated as the sample covariance of one-step teacher-forced predictor residuals over 256 training sub-trajectories of length 32, with a 10^{-6} diagonal floor. The observation map h is a ridge-regularised linear read-out from the state to the K taps (penalty 1.0), fitted on four fifths of the training encounters; the observation-noise covariance R is the diagonal of the residual variance of that map on the held-out fifth, grouped by encounter, with floor 10^{-6}. With ensemble mean $\bar{\mathbf{z}}_t^f = N^{-1}\sum_i \mathbf{z}_t^{f,(i)}$ and deviations $X_t^f = [\mathbf{z}_t^{f,(i)} - \bar{\mathbf{z}}_t^f]_i$, the update is the perturbed-observation analysis

$$P_t^f = \frac{X_t^f (X_t^f)^\top}{N-1}, \qquad K_t = P_t^f H^\top \left(H P_t^f H^\top + R\right)^{-1}, \tag{3.6}$$

$$\mathbf{z}_t^{a,(i)} = \mathbf{z}_t^{f,(i)} + K_t\left(\mathbf{y}_t + \boldsymbol{\epsilon}_t^{(i)} - H\mathbf{z}_t^{f,(i)}\right), \qquad \boldsymbol{\epsilon}_t^{(i)} \sim \mathcal{N}(0, R), \tag{3.7}$$

where H is the Jacobian of the linear h. Multiplicative inflation $\mathbf{z}^{f,(i)} \leftarrow \bar{\mathbf{z}}^f + \rho\,(\mathbf{z}^{f,(i)} - \bar{\mathbf{z}}^f)$ is applied before the update; ρ is selected per family on the validation split only ($\rho = 1.00$ for the predictive and linear families, $\rho = 1.05$ for the reconstructive one). The ensemble is initialised at the window start from a windowed pressure-to-state regressor (a Nyström radial-basis ridge with 300 components on a 12-frame causal pressure window) by sampling its training-residual covariance around the regressed mean, so no field information enters the filter at any time. The frozen observable probes that score the filter never enter the innovation: the only measured quantity in (3.7) is the K-tap wall pressure.

3.3.2. *The linear latent filter with encoded observations*

The structural-consistency retrofit, adapted from the latent-space assimilation construction of Tong, Wang and Yan (arXiv:2603.06752, 2026), replaces both F and h. The dynamics are linear, $F(\mathbf{z}) = A\mathbf{z}$, with A the ridge solution of $\mathbf{z}_{t+1} \approx A\mathbf{z}_t$ over all training transitions (penalty 1.0) followed by a projection of its singular values onto $[0, 1]$, the hard analogue of their spectral hinge penalty; Q is the covariance of the resulting fit residuals. The observation side replaces the sparse taps by an encoded full-state observation: a delay-embedding operator lifts the normalised tap history $\mathbf{p}_{t-m+1:t} \in \mathbb{R}^{Km}$ with $m = 10$ (causal, edge-padded), and a ridge map W (penalty 1.0) regresses it onto the state, so that

$$\mathbf{y}_t = E_{\text{obs}}(\mathbf{p}_{t-m+1:t}) = W^\top \operatorname{vec}(\mathbf{p}_{t-m+1:t}), \qquad h = \mathrm{Id}, \qquad R = \mathit{\Gamma}, \tag{3.8}$$

with $\mathit{\Gamma}$ the covariance of the E_{obs} training residuals plus a 10^{-8} jitter. The same E_{obs}, evaluated framewise with no dynamics at all, is the static delay-embedded inverse that anchors the estimator ladder. Filter divergence is declared by one fixed criterion everywhere in the paper: the normalised innovation squared exceeds the $\chi^2_{0.99}$ quantile at the observation dimension for five or more consecutive frames, or any analysis Mahalanobis ratio exceeds 3. The observation-rate protocol subsamples the update, not the sensor: pressure is recorded every frame and the analysis step is applied every m-th frame only.

3.3.3. *Forecast-derived state-dependent process noise*

The forecast-noise filter keeps the encoded observation (3.8) and replaces the member propagation by the direct multi-horizon forecaster of §3.2.1: each member is advanced by the one-step median forecast, and the process noise is read off the forecaster's own uncertainty band at forecast time,

$$\mathbf{z}_{t+1}^{f,(i)} = \hat{\mathbf{z}}_{t+1}^{q_{0.5}} + c\,\boldsymbol{\sigma}_{t+1} \odot \boldsymbol{\xi}^{(i)}, \qquad \boldsymbol{\sigma}_{t+1} = \frac{\hat{\mathbf{z}}_{t+1}^{q_{0.9}} - \hat{\mathbf{z}}_{t+1}^{q_{0.1}}}{2 \times 1.28155}, \qquad \boldsymbol{\xi}^{(i)} \sim \mathcal{N}(0, \mathrm{Id}), \tag{3.9}$$

so $Q_t = c^2 \operatorname{diag}(\boldsymbol{\sigma}_{t+1})^2$ is state-dependent through the predicted inter-quantile width (the denominator converts the $q_{0.9} - q_{0.1}$ band of a Gaussian to its standard deviation). The single scalar c is the filter's only tuned quantity and it carries two independent, validation-only calibrations. The production value $c = 1.77$ makes the one-step $q_{0.1}$–$q_{0.9}$ band cover 80 % of held-out training transitions. A second calibration, pre-registered

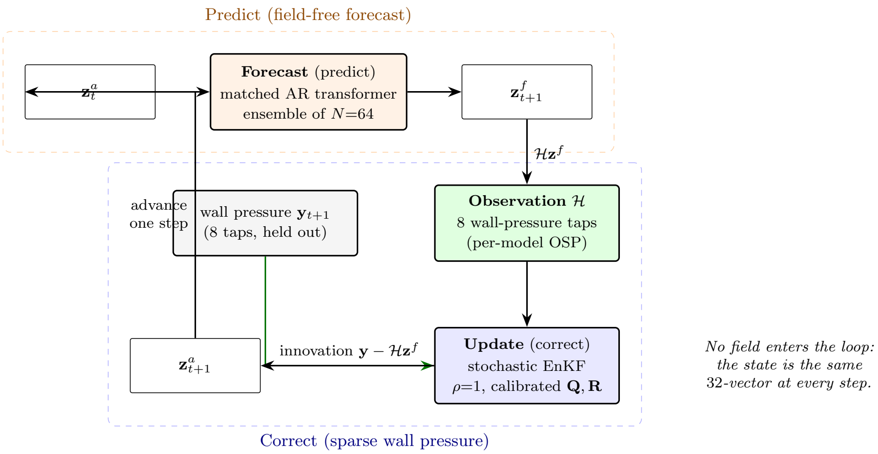


Figure 5. The pressure-only predict-correct estimator in the latent space. An ensemble of 64 members is advanced field free (predict), then corrected from 8 span-averaged wall-pressure taps (correct); the innovation is the difference between the measured observation and the observation predicted from the forecast state, in tap space for the base filter of §3.3.1 and in encoded latent space for the filters of §3.3.2 and §3.3.3. No vorticity field enters the loop and the lift and wake probes never enter the innovation. The state is the same 32-vector at every step.

Table 3. Configuration of the pressure-only ensemble Kalman filter in the latent space (§3.3, figure 5). Every setting is frozen before the held-out evaluation. The multiplicative inflation is selected per family on the validation split, since the families differ in dispersion: the predictive and linear filters select no inflation ($\rho = 1.00$), being under-dispersed, while the reconstruction filter selects $\rho = 1.05$.

Component	Choice
Filter	stochastic ensemble Kalman filter, latent space
Ensemble size N	64 members
Forecast model	matched autoregressive transformer, refit on frozen train trajectories
Observation operator $\mathcal{H}$	$\mathbf{z} \mapsto K$ span-averaged wall-pressure taps
Sensor count K	8 taps, per-model optimal (OSP) placement
Process noise $\mathbf{Q}$	from the predictor's one-step residual covariance
Observation noise $\mathbf{R}$	calibrated from held-out pressure residuals
Inflation ρ	validation-selected per family (1.00 predictive/linear, 1.05 reconstruction)
Initialisation	field-free: windowed pressure-to-latent regressor + its residual covariance
Assimilation window	[impact − 24, impact + 48] frames
Leakage guard	lift and wake probes never enter the innovation (enforced, unit-tested)

before any value was seen, selects c on the held-out encounters of the training cases by matching the pooled impact-phase normalised innovation squared, $\mathrm{NIS}_t = \nu_t^{\top} S_t^{-1} \nu_t / d$ with $\nu_t = \mathbf{y}_t - \bar{\mathbf{z}}_t^f$ and $S_t = P_t^f + \Gamma$, toward its consistency expectation of one, over the fixed grid $c \in \{1.0, 1.4, 1.77, 2.5, 3.5, 4.5, 6.0\}$, followed by a single frozen run on the test splits. The outcome of that calibration, reported in §4.4.3, is itself a result. The two-stage filter applies the encoded-observation update (3.8) with the median-forecast propagation (3.9) inside the impact window only, and outside it propagates members with the matched transformer and assimilates in tap space through the linear map of §3.3.1; the schedule boundary is the frozen impact-window mask of the cache and is not tuned.

Table 4. Per-estimator configuration (Gupta *et al.* 2026, table 1 analogue). All fits use the training split only; ρ and the two c calibrations use validation-family splits as stated in the text and nothing downstream of them. Ensemble size $N = 64$, taps $K = 8$ at the family's own placement, delay $m = 10$, every-frame assimilation, and the divergence criterion of §3.3.2 are common to all rows.

estimator	dynamics F	observation h	process noise Q_t	noise scale
static delay-embedded inverse	none	E_{obs}, eq. (3.8)	none	none
base EnKF (frozen)	matched transformer	linear $\mathbf{z} \to \mathbf{p}_K$	teacher-forced residual cov.	$\rho = 1.00$ / 1.05
linear latent filter	linear A (SV-clipped)	E_{obs}, $h = \mathrm{Id}$	A-fit residual cov.	$\rho = 1.0$
forecast-noise filter	forecaster median, eq. (3.9)	E_{obs}, $h = \mathrm{Id}$	quantile band, eq. (3.9)	$c = 1.77$ (coverage)
two-stage filter	median forecast inside impact, transformer outside	E_{obs} inside, tap space outside	as per stage	c as the forecast-noise filter
fixed-lag RTS	linear A	E_{obs}, $h = \mathrm{Id}$	A-fit residual cov.	lag $L = 5$

3.3.4. *Fixed-lag smoother*

On the linear stack of §3.3.2 a fixed-lag Rauch–Tung–Striebel smoother provides the reanalysis reference. With forecast and analysis moments $(\mathbf{z}_k^f, P_k^f)$ and $(\mathbf{z}_k^a, P_k^a)$ from the forward Kalman pass, the backward recursion over the lag window is

$$C_k = P_k^a A^\top \left(P_{k+1}^f\right)^{-1}, \qquad \mathbf{z}_k^s = \mathbf{z}_k^a + C_k \left(\mathbf{z}_{k+1}^s - \mathbf{z}_{k+1}^f\right), \tag{3.10}$$

applied over a lag of $L = 5$ frames, that is a fixed reporting latency of $0.25\, c/u_\infty$; the smoothed estimate at t uses pressure up to $t + L$ and nothing later. The ensemble smoother variant on the forecast-noise filter (a lagged cross-covariance update of the stored member history) degrades the filter and is reported as a negative in the supplementary material.

3.3.5. *Sensor placement*

Each family senses at its own taps. The per-family placement is a greedy forward selection over the 192 physical wall taps: at each step the candidate tap whose 30-frame causal pressure window, concatenated to the already-selected windows, best ridge-predicts the leading principal component of the family's own impact-frame states is added. Writing r_j for the ridge residual mean square of the joint feature set extended by tap j, and $g_j = \max\{0,\, r_\varnothing - r_j\}$ for its reduction over the target variance, the selected tap maximises

$$J(j) = n\, g_j - \tfrac{1}{2}\, g_j - \tfrac{1}{2}\, n\, r_j + \tfrac{1}{2}\, \frac{n\, g_j}{g_j + 10^{-6}}, \tag{3.11}$$

with n the number of training encounters; at this configuration the first and third terms dominate, so the criterion is explained-variance gain penalised by the residual. The staircase is nested by construction, each budget warm-starts from the previous selection, so the $K = 16$ array contains the $K = 8$ array. The shared target-blind reference array is the column-pivoted QR (qDEIM) selection of Drmač & Gugercin (2016) on the leading pressure modes. Which placement a given figure uses is stated in its caption.

The mixed timescales of this estimator, a transition map trained on sequences and an emission map trained per instant, mirror the structure of the training objective in §3.1: a dynamics equation is a statement about trajectories and an observation equation is a statement about single instants. The combined loss is a jointly trained state-space realisation, and the filter is the same decomposition read at inference.

3.4. *Diagnostics, controls, and uncertainty*

Two diagnostics turn the closure result into a mechanism. The first is the latent drift, which addresses the fact that a predictor with low one-step error need not produce a useful long rollout, because each prediction is fed back as the next input and small geometric errors accumulate. We quantify how far a rollout has left the training manifold

Table 5. The compared families: encoder, latent, and training objective. Every family produces a pooled $d = 32$ coefficient vector at the same dimension. The top block is the controlled matrix, one shared encoder (the hybrid convolutional-stem plus vision transformer of table 2) with the training ingredient varied one at a time; the objective-free supervised control trains that encoder with the observable heads and the anti-collapse term alone, with no predictor and no decoder, so it isolates what the supervision by itself supplies. The bottom block is the published reference recipes: the Fukami autoencoder and the β-VAE are plain convolutional autoencoders, a convolutional stack down to a fully-connected bottleneck with no skip connections, and POD is a linear basis. The forecast merit of table 6 reads each latent through a separate matched forecast operator; the operators (an autoregressive transformer and a residual U-Net) and the reason a different one is stable on the reference latents are described there and in Appendix A.

Family	Encoder	Latent	Training objective	Source
predictive	CNN + ViT (shared)	pooled 32-vec	latent-space prediction + heads + SIGReg	this work
supervised only	CNN + ViT (shared)	pooled 32-vec	heads + SIGReg only (no objective)	this work
reconstructive AE	CNN + ViT (shared)	pooled 32-vec	field reconstruction + heads + SIGReg	this work
anti-collapse AE	CNN + ViT (shared)	pooled 32-vec	field reconstruction + SIGReg (no heads)	this work
Fukami AE	conv. AE, no skips	pooled 32-vec	reconstruction + observable head	Fukami & Taira (2023, 2025)
β-VAE	conv. VAE, no skips	32-vec (mean)	reconstruction + KL + heads	Solera-Rico *et al.* (2024)
POD	linear projection	32 leading modes	none (SVD basis)	classical

by the Mahalanobis distance of the rolled-out latent at horizon H, measured against the distribution of simulation-encoded latents for the same encoder, normalised by the Mahalanobis distance of the simulation-encoded latent itself. A ratio near one means the rollout stays within the natural spread of the training latents; a large ratio means it has gone out of distribution, so that the probe is then queried outside its support.

The second diagnostic is a parameter-only floor, which carries particular force here because the model never sees the gust parameters: it bounds what the parameters alone could explain, so that any closure above it is genuinely supplied by the latent state. We compute it with a kernel-ridge regressor that maps $\mathbf{c} = (G, D, Y)$ directly to each observable at $H = 16$, with no latent input, fit on training and evaluated on the held-out splits. On wake enstrophy this floor is low, so the latent's wake closure is not something the parameters could have supplied. At the impact frame the same parameter floor is high, because the just-released gust nearly determines the instantaneous state; the gap opens as the wake evolves (§4.1.3).

Because the predictive encoder differs from the reconstructive baseline in objective, architecture, and auxiliary supervision at once, the comparison is accompanied by controls that vary one factor at a time (table 5): a crossing of objective with architecture at matched auxiliary heads, an ablation of each auxiliary head, and a conditioned predictor variant that adds $\mathbf{c}$ (§4.2.4). All the controls share the one encoder of table 2 and differ only in what is trained on top of it. The objective-free supervised control is the sharpest of these: it trains the shared encoder with the wake and lift heads and the anti-collapse term but with no predictive or reconstructive objective, no predictor and no decoder, so it holds the supervision fixed and removes every objective, isolating what the observable heads alone supply. The published reference recipes, a plain convolutional autoencoder in the Fukami–Taira lineage (Fukami & Taira 2023, 2025), the β-variational autoencoder of Solera-Rico *et al.* (2024), and a linear proper-orthogonal-decomposition basis, use their own native architectures but produce a pooled coefficient vector at the same latent dimension. The forecast merit refits a matched operator on each frozen latent; we use the operator that is numerically stable on it, a transformer on the pooled states and a residual U-Net on the reference latents, on which the transformer diverges (§3.5, Appendix A).

Every held-out R^2 and mean absolute error reported below carries three independent uncertainty signals, following the reporting protocol fixed with the data split: a 2000-

resample bootstrap over the held-out encounters, the standard deviation across three encoder retrains from independent seeds, and a five-fold cross-validation over cases on the probe read-out. Their full specification is in Appendix A.

3.5. *Protocol, phases and uncertainty accounting*

Every encounter is scored in three phases anchored to its impact frame: lead-in $[t_{\mathrm{imp}} - 8, t_{\mathrm{imp}})$, impact $[t_{\mathrm{imp}}, t_{\mathrm{imp}} + 16)$ and relaxation $[t_{\mathrm{imp}} + 16, t_{\mathrm{imp}} + 48)$, all half-open in frames and clamped to the encounter. Load metrics are reported in three complementary forms because each corrects a failure mode of the others: the coefficient of determination $R^2 = 1 - \mathrm{SSE}/\mathrm{SST}$ pooled over the phase frames, the root-mean-square and mean absolute errors in lift-coefficient units, and peak-relative quantities. The peak diagnostics locate the extremum of $|C_L|$ within the impact phase in the estimate and in the truth separately and report the value difference, its magnitude as a percentage of the true peak, and the timing difference in convective units at $\Delta t = 0.05$. Peak-region readability uses a pooled R^2 over windows of half-width 8 frames centred on each encounter's true peak, with a persistence floor that holds the mean of frames $[0, 25)$ constant; phase lag is the argmax of the normalised cross-correlation within ± 20 frames with parabolic sub-frame refinement, positive when the estimate trails the truth. Decoded fields are scored by the structural similarity index in the convention of Wang *et al.* (2004), $K_1 = 0.01$, $K_2 = 0.03$, with the data range pinned per dataset version to twice the global 99.9th percentile of the normalised target magnitude ($L = 8.487$ for the present split), on three masks: the feathered near-body band of §3.1.1, a wake window spanning $0 < x/c < 4.5$ and $|y/c| < 1.25$, and the full frame. Linear-probe R^2 is used throughout as a readability metric and never as an information metric; the control that motivates this distinction, a multilayer probe that equalises what a linear read-out cannot reach, is reported alongside every dimension comparison (§4.1.2).

Uncertainty is accounted at three levels and every stochastic number in the paper carries its count. Encoder training variance uses three seeds wherever a cell is compared (the conditioning cube, the $d = 4$ bands, the direct-predictor and reconstruction-baseline bands); single-seed cells appear only in the dimension grid and are disclosed as such in each caption. Filter member-noise variance uses five seeds of the perturbed observations and initial ensemble. Paired family comparisons use a case-clustered bootstrap that resamples whole cases with replacement (2000 resamples on encounter means, 10^4 on case means) and reports the interval that excludes or straddles zero; multiple comparisons within one question family are Holm-corrected. The acceptance gates for the conditioning study and for the present campaign's follow-up runs were written and committed before any corresponding result was read, and both pre-registrations ship with the data record. The gust-intensity axis is reported in the inventory sign convention, and no test-C quantity is used for any selection anywhere in the pipeline.

4. Results

We report the controlled representational comparisons (§4.1 to §4.3) on the in-distribution test encounters and hold the boundary test set in reserve, so that the $|G| = 4$ extrapolation enters only as a final reported number and never as a selection signal. The estimator, the recovery, and the operating envelope (§4.4) are characterised with a frozen filter across all encounters, training, validation, in-distribution test and boundary test sets alike, and the per-encounter paired tests on the in-distribution test and boundary test sets are given in Appendix A. Every quantity quoted below is a held-out value at the readout horizon unless stated otherwise. One division of labour is declared once and holds throughout:

the wake-supervised predictive state is the estimation-thesis state that the assimilation grid and the operating envelope run on, and the lift-focused predictive state is the conditioning result whose accuracy ceiling parts B and C quantify.

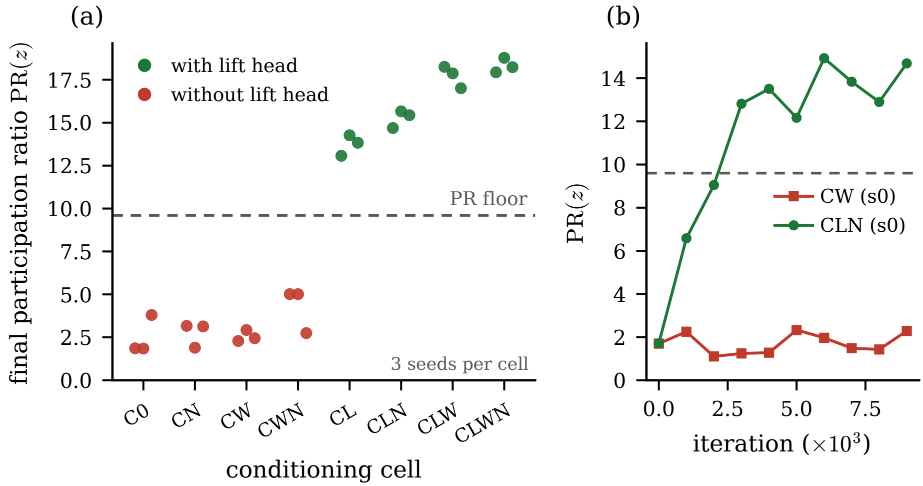
4.1. *Constructing a physically useful state*

4.1.1. *Constructing the state: which supervision buys which property*

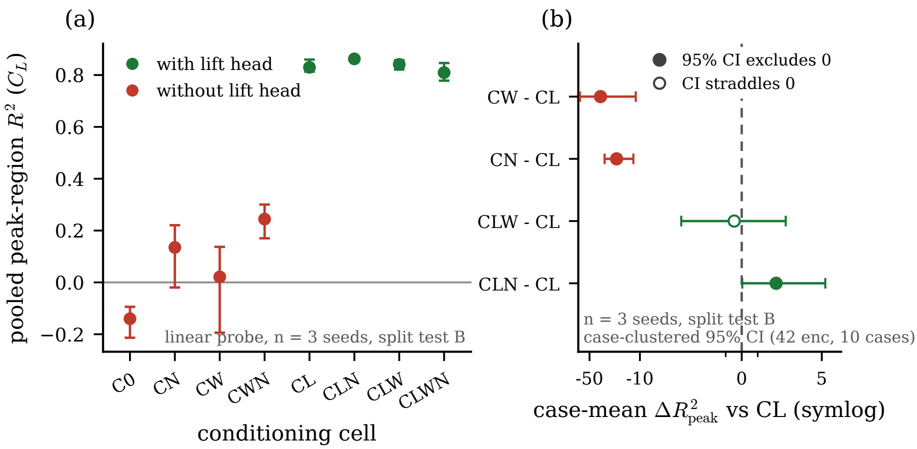
The construction question is settled by a $2 \times 2 \times 2$ cube over the three supervision heads $\{L, W, N\}$, trained at three seeds per cell on the pooled pipeline of §3.1 with matched reconstructive anchors, under gates written and archived before any result was read; probes are fit on train and scored on the held-out in-distribution test encounters, the validation set steering nothing and the boundary test set staying in reserve per the policy above. The central result is structural: under the predictive objective the scalar lift head is load-bearing for latent health. Every cell without L collapses, with participation ratios between 1.05 (C0) and 1.00 (CWN) against a floor of 9.6, flat from the first diagnostic; neither the wake observable, nor the near-body observable, nor both together substitute for the scalar anchor (figure 6a). The reconstruction objective anchors without it: the wake-only autoencoder keeps a healthy latent (participation ratio 1.00) yet reads the peak loads poorly ($R^2 = 0.47$), so a healthy latent and a lift-readable latent are different achievements. The collapse is the low-intrinsic-dimension failure mode anticipated for the characteristic-function regulariser at kit strength; the automatic variance-covariance fallback is armed at an iteration threshold the pooled runs never reach, so the cube exposes the mechanism rather than masking it.

Nothing replaces the lift anchor, and the near-body head is the one conditioning that adds to it. Replacing L by W is refuted catastrophically: the wake-only cell trails the lift-only cell by -35.22 peak-region R^2 points (case-clustered interval [-67.97, -11.42]), and replacing L by N fails the same way (-21.06 [-30.91, -12.33]). On top of L, however, the Chang head buys real lift accuracy: the lift-focused predictive state (cube code CLN) beats the lift-only cell CL by 2.15 points with an interval that excludes zero ([0.03, 5.59]), and is the strongest cell in the cube, with pooled peak-region $R^2 = 0.86$ at the tightest seed band anywhere and a decoded phase lag of -0.011 t/c (figure 6b). The wake head's marginal value on top of L is geometry, not lift: the wake-supervised predictive state (cube code CLW) matches CL on the peaks (interval [-3.78, 2.75] straddles zero) while adding near-body structural fidelity, SSIM in the Wang convention of §3.5 (0.017 [0.009, 0.026] SSIM).

The three heads therefore divide the labour: L anchors the latent, N sharpens the load, and W carries the wake state. The wake-enstrophy probe makes the last division quantitative: the wake-supervised predictive state reads E_w at $R^2 = 0.73$ linear and 0.84 through a multilayer probe (three seeds, in-distribution test set), where the lift-focused predictive state reaches only 0.37 and 0.63. Honesty requires the anchor comparison too: the reconstructive AE-LW carries the most linearly readable wake state of all (0.77 linear, 0.90 MLP), as a reconstruction objective should; the predictive family's wake advantage is not probe readability but what survives forecasting and assimilation, quantified in §4.2.3 and §4.4.3. The full task-dependent readability matrix, five observables across every cell and anchor, is figure 7: the wake block lights up exactly where wake supervision or a reconstruction objective sits, and nowhere else. One circularity deserves preemption: supervising on lift and winning on lift would be circular if the anchor claim were an accuracy claim. It is not. The anchor claim is about latent health, established by the collapse pattern of the no-L cells; the accuracy claim is the increment of N over an already lift-supervised baseline, established under paired pre-registered gates. The two claims rest on different comparisons and fail independently.



(a)



(b)

Figure 6. The conditioning cube (in-distribution test set, three encoder seeds per cell). (a) Final participation ratio per cell against the collapse floor $0.3\,d = 9.6$: every no-L predictive cell collapses, every L cell is healthy; the inset shows the collapse is flat from the start of training while a healthy cell crosses the floor within the first quarter of the run. (b) Pooled peak-region lift R^2 per cell (left) and paired case-mean deltas against the lift-only cell CL with case-clustered 95 % bootstrap intervals over the 42 encounters of 10 cases (right, symmetric-log axis); filled markers mark intervals that exclude zero. The lift-focused predictive state cell (CLN) is the unique positive increment.

4.1.2. *The decode floor*

Decoding closes the loop on the construction story: the same division of labour that the probes read appears in the decoded fields, scored on the in-distribution test set through identical-capacity frozen decoders with the Wang SSIM convention of §3.5. With the lift anchor present, the near-body head buys the same near-body structural fidelity as the wake head (0.709 for the lift-focused predictive state against 0.709 for the wake-supervised predictive state), the collapsed no-L cells sit far below (from 0.470 for CW), and the reconstructive anchors reach parity (0.704 for AE-LW), as an objective trained on the field should (figures 8 and the per-cell bars in the data record). The decode panels make the

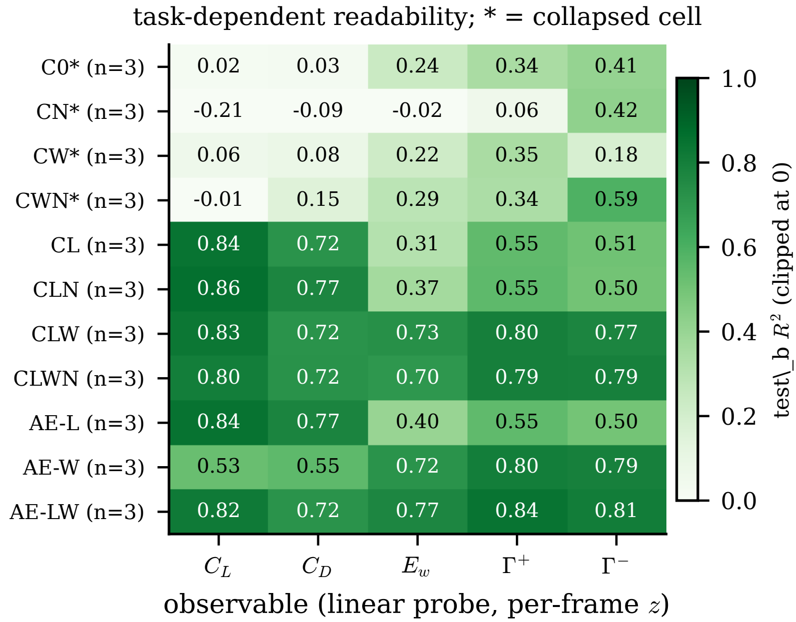


Figure 7. Task-dependent readability (in-distribution test set, linear probes on the per-frame state, seed mean over three encoder seeds per cell): lift and drag follow the L head, the wake block (E_w, $\Gamma^{\pm}$) follows the W head or the reconstruction objective, and collapsed cells (asterisks) read nothing linearly.

same point qualitatively: all three healthy states recover the near-body vortex system and the coherent wake at matched fidelity, and the residual rows concentrate on the fine-grain shed structures no 32-coefficient state carries.

4.1.3. *What the coefficient state carries, and what supplies it*

The comparison between reduced states is meaningful only if the wake is a discriminating endpoint, and it is. At the readout horizon the release parameters no longer fix the wake enstrophy, while they still fix the forces, so the wake is where a representation that has captured the gust-vortex physics parts company from one that has captured only its force signature. Table 6 reports what each pooled coefficient state carries, read by a fixed linear probe from a held-out in-distribution test field with no rollout. The wake is read cleanly by every state that carries the wake-supervision head, with a held-out R^2 between 0.750 and 0.792, and it collapses for every state that does not, falling to 0.160 for the predictive latent trained without the head. Instantaneous readability of the wake is therefore supplied by the observable supervision, not by any training objective.

The objective-free supervised encoder makes this concrete. Trained with the wake head but no predictive or reconstructive objective at all, it reads the wake at least as well as the full predictive latent (0.792 against 0.751): the paired readability gap is 0.041 in the wake with a confidence interval $[-0.047, 0.162]$ that includes zero, so the instantaneous readability is supplied by the supervision, not by the training objective. Under one shared forecast operator, the direct multi-horizon forecaster of §3.2.1 trained identically on every state, the three wake-headed states are also the most forecastable and are statistically indistinguishable from one another (0.561, 0.574 and 0.443 at horizon sixteen, case-clustered intervals overlapping), so the forecastability of the coefficient state, like its readability, is supplied by the supervision rather than by the training objective. What the predictive objective adds is the model it co-trains: the wake-supervised predictive state's

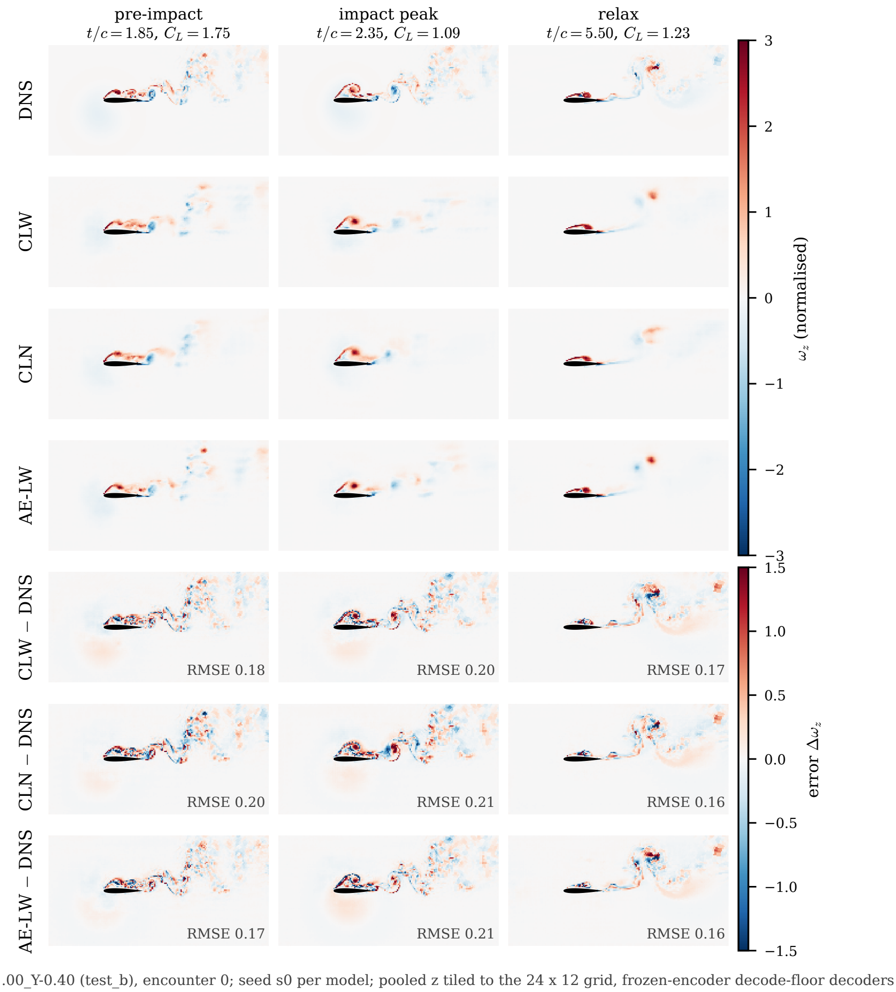


Figure 8. Decode floor on the representative in-distribution test encounter (case G−0.50_D1.00_Y−0.40, encounter 0; one decoder seed per model): DNS, the three healthy states, and the error rows at three phases. Identical-capacity frozen decoders; normalised vorticity at the fixed ±3 range.

own vector predictor forecasts the five observables at 0.755 at horizon eight, above what the shared operator extracts from any latent there (0.680), and is overtaken by the direct multi-horizon forecaster at horizon sixteen (0.418 against 0.561), the error-accumulation contrast that §4.2.3 quantifies. We state the representational result as the supervision's, and turn to what the predictive objective does add, which is dynamical.

One inversion in the table fixes the coordinate-level reading. The predictive latent trained without the wake head carries the highest single-observable lift forecast of any state (0.691) yet the lowest wake readability (0.160), because the forces are held redundantly across the latent while the wake is not, the distributed-code signature we return to in §4.1.4. Among the reference recipes the lift-augmented reconstructive autoencoder reads the wake only when it carries the wake head (0.432 against −0.094 without it), and the linear basis does not reach the wake at all (0.186), so no linear projection recovers the enstrophy that a supervised nonlinear state does.

Table 6. What each pooled coefficient state carries, at matched dimension $d = 32$ on the 42 in-distribution test encounters. Held-out wake-enstrophy readability (fixed linear probe, no rollout); forecast merit under one shared operator, the direct multi-horizon forecaster of §3.2.1 trained identically on every state (three operator seeds; mean over the five observables of the pooled nonlinear-probe R^2 at horizon sixteen; seed means, with case-clustered bootstrap intervals in the text); field-reconstruction error (VRMSE at horizon eight); and decode similarity (SSIM). The controlled matrix (top block) moves one training ingredient at a time; the reference recipes (bottom block) use their own architectures (table 5). Wake readability separates on the wake-supervision head, not on the training objective, and under the shared operator the three wake-headed states are also the most forecastable, statistically indistinguishably from one another. The wake-supervised predictive state's own co-trained predictor leads every fitted operator at short horizon (0.755 at horizon eight) and is overtaken by the direct multi-horizon forecaster at horizon sixteen (0.418 against 0.561), the error-accumulation contrast of §4.2.3.

State	Objective	Wake head	wake R^2	merit	field VRMSE	SSIM
predictive (wake)	predictive	yes	0.751	0.561	0.962	0.779
supervised only	none	yes	0.792	0.443	0.992	0.774
AE (wake)	reconstruction	yes	0.750	0.574	0.976	0.776
predictive (no wake)	predictive	no	0.160	−0.426	0.983	0.767
AE (no wake)	reconstruction	no	0.456	0.285	0.976	0.773
anti-collapse AE	reconstruction	no	0.333	0.068	0.949	0.782
β-VAE	reconstruction	yes	0.728	0.040	1.007	0.724
Fukami	reconstruction	no	−0.094	−0.195	1.016	0.708
Fukami (wake)	reconstruction	yes	0.432	0.162	1.026	0.705
POD	linear	no	0.186	0.036	0.930	0.775

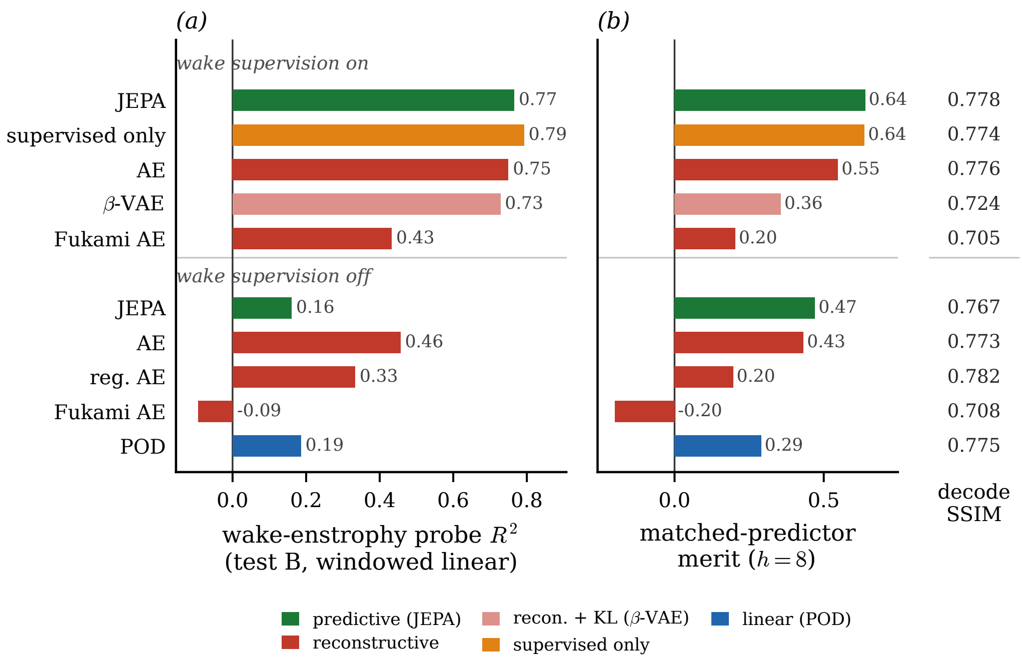


Figure 9. The attribution result of table 6, at a glance. (a) Held-out wake-enstrophy readability and (b) matched-predictor merit, with the models grouped by whether they carry the wake-supervision head; every wake-headed state reads the wake and every bare state collapses, whatever the training objective. The right-hand column lists the decode similarity, which barely separates the families.

4.1.4. *The physics the state holds*

The estimation results rest on a representation, and this subsection reads what the representation holds in measures that do not depend on the filter. The predictive latent carries the shedding clock. Fitting a linear operator to the encoded limit-cycle trajectory by dynamic mode decomposition places a marginally stable oscillatory mode at Strouhal number 0.666, against the measured shedding line, with modulus 0.993 just below unity. The Fukami-lineage autoencoders lose that clock: their leading latent mode is heavily damped and off-frequency (0.182 at modulus 0.590 without the wake head, 0.147 at 0.294 with it). The structured states keep it, whether their structure comes from prediction, from supervision, from anti-collapse, or from a linear basis: the objective-free supervised latent, the anti-collapse reconstruction, and the proper orthogonal basis all place the shedding mode near the measured frequency on a near-unit-modulus orbit (0.654, 0.676 and 0.673 at moduli 0.989, 0.995 and 0.996). It is the Fukami-lineage reconstruction, not reconstruction in general, that fails to hold the shedding spectrum.

The parametric manifold coheres with the release parameters where the training data resolve them. An impact-frame probe of the predictive latent recovers the gust strength and the core diameter well (0.88 and 0.73 on the in-distribution test set), which is the atlas of figure 10 coloured by parameter. The wall-normal offset, unreadable at the linear-probe level in earlier work, becomes weakly readable here (0.29 linearly and 0.79 with a nonlinear probe), plausibly because the additional cases sample it more densely near the training centre; we note this rather than lean on it, since the offset remains the least resolved parameter (§2.2).

4.2. *Compression and forecastability*

4.2.1. *How small the state can be*

The dimension axis splits into two tiers, and the split is the paper's compression statement (figure 11a). At $d = 4$ the predictive families read the peak loads through a linear probe at 0.903 (wake-supervised predictive state lineage) and 0.910 (coupled-objective arm) against 0.796 for the reconstructive baseline, three seeds each with non-overlapping ranges; the lift-focused predictive state lineage trained through the direct multi-horizon forecaster is close to dimension-insensitive on the peaks, from 0.900 at $d = 4$ to 0.880 at $d = 32$ (three seeds at both ends), where the wake-supervised predictive state lineage declines with d (from 0.902 to 0.841). A four-coefficient lift-critical state is therefore viable. The wake state is not compressible to it: the decoded structural fidelity of the same lineage rises monotonically with dimension (full-frame SSIM 0.733 to 0.781, near-body 0.540 to 0.677), so the wake-bearing state needs $d \geq 16$ while the lift saturates by $d \approx 4$ to 8.

The probe-dilution control disciplines how the first tier may be read (figure 11b). A multilayer probe nearly equalises every dimension (from 0.89 at $d = 4$ to 0.88 at $d = 32$ on the in-distribution test set): the lift information in the state is dimension-invariant, and small states do not know more. What changes with dimension is linear accessibility. The best four coordinates of a larger state recover only 0.66 to 0.55 through a linear read-out, so the code is distributed rather than diluted, and the $d = 4$ advantage is exactly that four coordinates are what a linear probe, a filter gain, or a controller reads directly. The honest form of the two-tier claim is therefore: compact and linear at $d = 4$, distributed and partly nonlinear above.

4.2.2. *The distributed wake code*

The wake is a collective code rather than a labelled coordinate. The gap between the wake readability of the full predictive latent and that of its single most informative coordinate

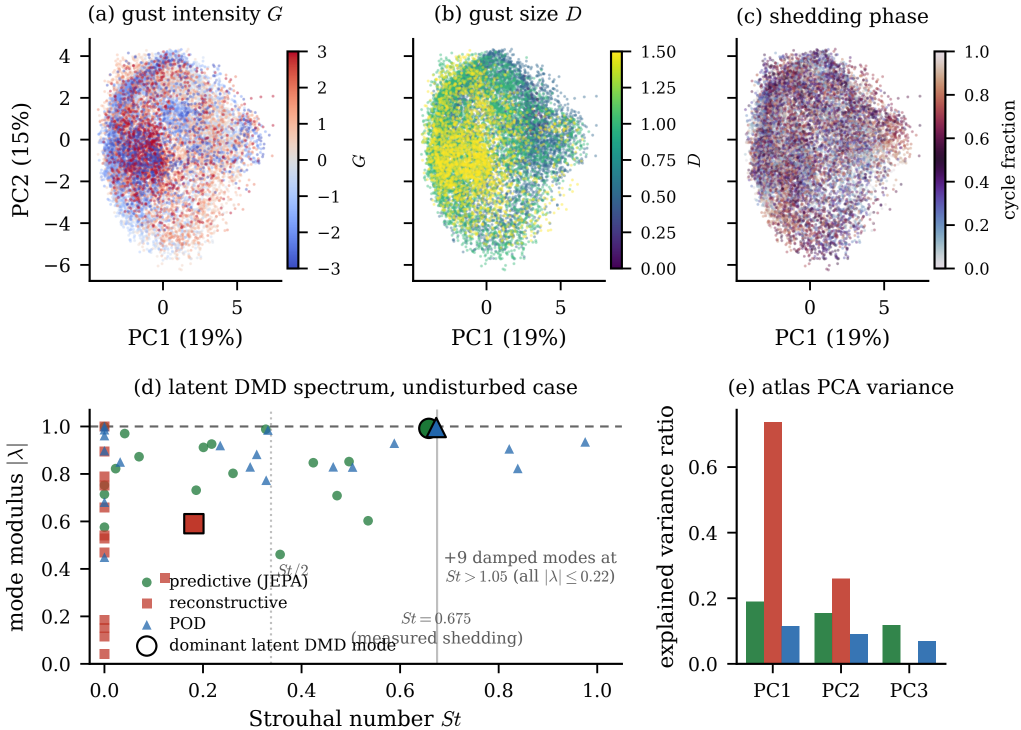


Figure 10. The physics the latent state holds. Top: two-dimensional projections of the predictive coefficient latent coloured by gust strength, core diameter, and shedding phase, showing a parametric manifold that coheres with the release parameters. Bottom: the latent dynamic-mode-decomposition spectrum, modulus against Strouhal number, by family; the predictive and linear latents place a marginally stable mode at the measured shedding frequency while the Fukami-lineage reconstruction is damped and off-frequency.

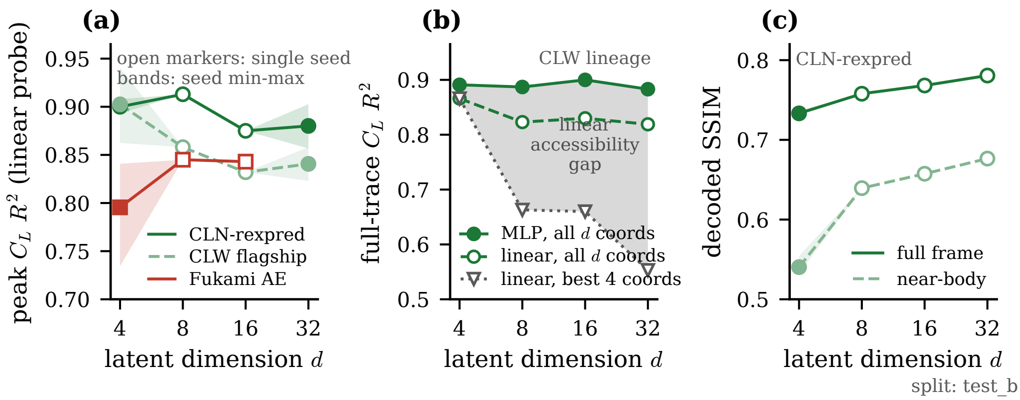


Figure 11. The dimension axis on the in-distribution test set. (a) Peak-region lift R^2 (linear probe) against latent dimension for the three lineages; shaded bands are three-seed min-max ranges, open markers single seeds. (b) The probe-dilution control: a multilayer probe is flat in d while the best-four-coordinate linear probe falls, so the $d = 4$ advantage is linear accessibility of a distributed code, not information. (c) Decoded SSIM (Wang convention, §3.5) of the same lineage rises with d: the wake tier is not compressible.

is 0.51, and 0.53 for the objective-free supervised state, against −0.03 and 0.04 for the reconstructive and linear baselines, so no single predictive coordinate carries the wake and the information is spread across many. The dimension at which each family first reaches a wake R^2 of one half makes the same point from the other side: the predictive and the wake-headed reconstructive states need eight latent dimensions (8 each), while the linear basis never reaches it at any dimension we tested. The headline dimension of thirty-two sits on a plateau rather than a cliff, the wake readability moving only 0.053 across sixteen, thirty-two and sixty-four (0.751 at thirty-two), so the choice of dimension is a robustness statement and not a tuned operating point.

A spatially resolved latent decodes the vorticity field more sharply than this coefficient state, but it is of far higher dimension than the wall measurement and so is not recoverable from sparse pressure. That decodability-versus-estimability trade is the reason the coefficient state, not the field-resolved latent, is the object this paper estimates; we take it up in §5.3 and quantify its cost in the supplementary material.

4.2.3. *Advancing the state: forecasting without an oracle*

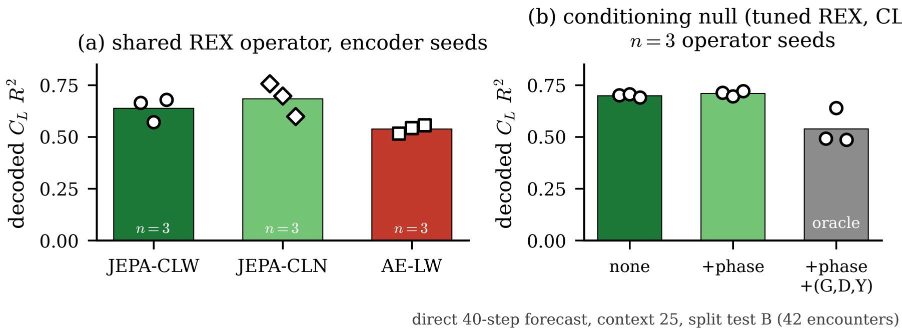
The family ordering survives an operator-robust protocol, which resolves the operator confound the suited-operator comparison of §3.2 left open: one direct multi-horizon forecaster (§3.2.1), trained identically on each family's frozen latents, is stable on every latent including the reconstructive ones. Under it the predictive states are the more forecastable, decoded lift $R^2 = 0.638$ for the wake-supervised predictive state and 0.685 for the lift-focused predictive state against 0.539 for the reconstructive anchor, three operator-training seeds per family on the in-distribution test set, with the lift-focused predictive state latents the most forecastable of all (figure 12a). The ordering agrees with the suited-operator protocol, so it is a property of the states, not of the operator that reads them.

The forecasting mechanism is the direct-versus-autoregressive contrast. Rolled autoregressively through the impact, the matched transformer compounds its own errors: forty steps from the pre-impact context its decoded lift closes at -0.62 on the wake-supervised predictive state latents and -0.23 on the lift-focused predictive state latents. The direct multi-horizon forecaster, emitting all forty steps in one shot from the same context, closes at 0.701 on the identical protocol (figure 12b): rollout compounding, not the state, is the failure mode. The direct forecast is a conditional median, and the honest reading is that of a median: it contracts toward the climatology of the shedding cycle where the encounter is unresolved, which is precisely the model-error regime the assimilation of §4.4.3 must correct.

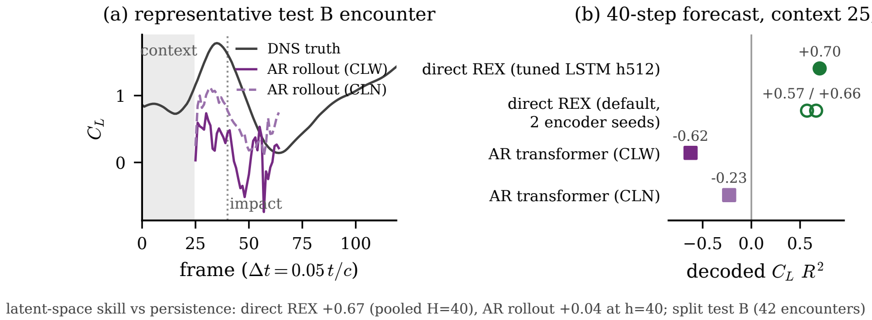
Knowing the gust does not help. Conditioning the forecaster on the true episode parameters (G, D, Y), an oracle no deployment has, degrades the forecast in every one of three operator seeds, 0.539 against 0.699 unconditioned, with non-overlapping seed bands: at the present case count the oracle covariates overfit held-out parameter combinations. A deployable phase covariate is a wash within seed noise (0.710). This closes the estimation thesis from the third side: forecast alone contracts, parameter oracles hurt, and what remains is the model-plus-pressure route of part D.

4.2.4. *Why the state stays usable under rollout*

A state that reads the wake at a single instant need not survive a rollout, because a probe attached to a drifted latent is queried outside its support. We ask where each rolled state goes relative to the training distribution by decomposing its departure onto the principal and the near-null directions of the encoded covariance (table 7). The reconstruction carrying only an anti-collapse term is near-isotropic, its covariance condition number close to unity (11), and its rollout leaks a tenth of its departure into the near-null directions (0.100).



(a)



(b)

Figure 12. Forecasting the state on the in-distribution test set. (a) Decoded lift R^2 of the shared direct multi-horizon forecaster per family (three operator seeds each, bars at the seed mean) and the conditioning null (three seeds per arm): the (G, D, Y) oracle degrades the forecast, the phase covariate is a wash. (b) Direct versus autoregressive over forty steps through the impact from a 25-frame context: the one-shot forecast stays usable where the rollout compounds.

The supervised states, predictive and objective-free alike, are strongly anisotropic, their variance concentrated in a low-rank subspace (condition numbers 451 and 1041), and their rollouts keep their departures inside it (0.014 and 0.029, against the 0.25 of an isotropic null). Supervision, not the anti-collapse term alone, builds the protected subspace: it pins the encoded variance onto the observable-aligned directions, and the forward model, having only ever seen states there, moves within it. The anti-collapse term on its own drives the latent toward isotropy, as an independent adaptation of the same regulariser to biosignals also reports (Broustail *et al.* 2026), and it is the supervision that supplies the geometry the rollout needs.

What the predictive objective adds is long-horizon stability, and it is the multi-step form of the objective that adds it. Trained with a single-step rather than a multi-step rollout, the same predictive latent forecasts the five observables less well at every horizon, and the margin widens with the rollout: by sixteen steps the single-step latent's merit has fallen to 0.237 while the multi-step latent holds 0.488, with markedly lower on-manifold drift (0.518 against 0.733 at horizon sixteen). The single-step latent is marginally better on the load alone a few steps out (0.685 against 0.664 in C_L closure at eight steps), a single-observable exception to a forecast advantage that otherwise grows with horizon. The objective-free supervised state, which has the readability and the protected subspace but no predictive training, is the weakest field forecaster of the supervised group. The contribution of the predictive objective is therefore forward rather than instantaneous: it keeps the rolled state

Table 7. Where each rolled state goes relative to the training distribution, on the in-distribution test set at rollout horizon sixteen. The latent covariance condition number measures anisotropy, and the near-null departure fraction is the share of the rollout departure that falls into the bottom-quartile directions of the encoded covariance. An isotropic null would leak 0.25. The anti-collapse reconstruction is near-isotropic and leaks; the supervised states are anisotropic and stay in their low-rank subspace.

State	covariance condition number	near-null departure fraction
anti-collapse AE	11	0.100
predictive (wake)	451	0.014
supervised only	1041	0.029

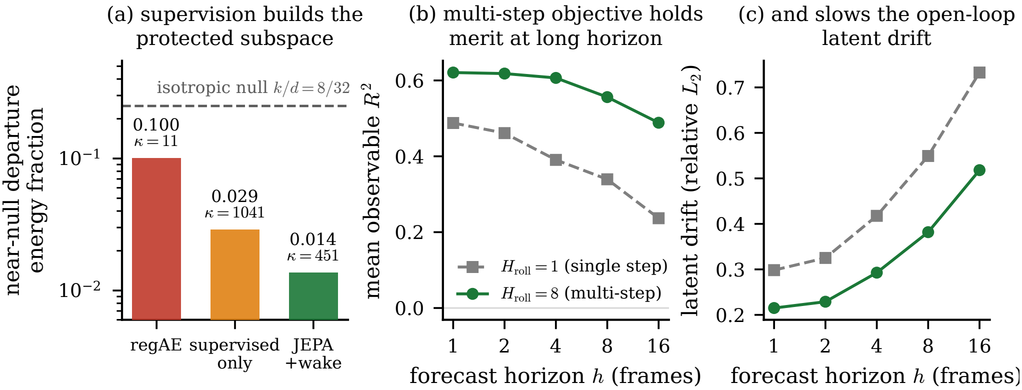


Figure 13. The rollout mechanism and what the multi-step objective adds. (a) Near-null departure fraction by family against the isotropic null, annotated with each latent covariance condition number: supervision builds the protected subspace. (b) Forecast merit and (c) rollout drift against horizon for the single-step and multi-step objectives, tied at one step and separated by sixteen.

both in distribution and near the true trajectory, which is exactly what a sequential estimator queries.

4.3. *What wall pressure observes*

At deployment the state must be read from the wall, so we ask how much of each latent a sparse set of pressure taps recovers over a window ending at the readout instant (table 8). Two findings organise the answer. First, raw-variance recoverability and physics recoverability disagree: the reconstruction latent is the most recoverable in state variance (0.921) yet carries the least wall-recoverable physics (0.470 in observable closure), while the predictive latent carries the most physics (0.589) from a less recoverable state (0.659 ± 0.009 over three seeds). The wall aligns with many of the reconstruction latent's directions, but they are the low-variance ones that do not carry the observables; it aligns with fewer of the predictive latent's directions, but those are the ones that hold the state. Second, perturbing the predictive state along its principal directions and measuring the signal that reaches the pressure head shows the wall reading the force and wake-enstrophy directions and blind to the highest-variance wake-circulation direction, a ranking unchanged under a fixed-amplitude perturbation and so a property of the learned dynamics rather than of the normalisation. The wall, in short, reads the force well and the wake circulation not at all, which is why the state must be estimated sequentially rather than inverted from one frame.

The margin of the coefficient state over a raw high-dimensional field latent is real but modest. Its recoverability advantage is 0.120 in R^2, with a confidence interval

Table 8. What eight wall-pressure taps recover of each family's coefficient state, on the in-distribution test set, with the recovery estimator selected by cross-validation per family and the observables read through the frozen probes. Raw-variance recoverability and physics recoverability disagree: the reconstruction latent is the most recoverable in state variance but the least in physics, while the predictive latent carries the most wall-recoverable physics from a less recoverable state. The absolute state recovery is estimator-limited: a per-family tuned LSTM (hidden width validation-selected up to 512) recovers more from the same taps, raising the predictive state to 0.833 and the reconstruction to 0.937, and it had not saturated at the largest network tested, so these are a lower bound on the wall-recoverable state; the family ordering, which is the claim, is unchanged.

State	recovery estimator	state R^2	observable R^2
predictive	LSTM	0.659	0.589
reconstructive (Fukami)	LSTM	0.921	0.470
linear (POD)	MLP	0.561	0.534

$[0.096, 0.145]$ that clears zero. We therefore do not claim a strong-form estimability gate; the defensible statement is the clean physics-versus-variance ordering above.

4.3.1. *Sensors traded for delays*

That the wall is blind to the wake circulation at a single instant is a statement about one frame, not about the sensor count, and the delay-embedding reading of §3.3 makes a testable prediction: the circulation the instantaneous pressure misses should become recoverable as the pressure is read over a longer window. It does. Holding the tap count at eight and lengthening the window, the recovery of the wall-blind wake circulation rises from near its single-frame value to 0.33 at a thirty-frame window, a gain of 0.224 with a case-clustered interval $[0.136, 0.343]$ that excludes zero, while the force, already visible at one instant, barely moves. The delays recover the coordinate the sensors cannot.

Varying the tap count and the window together traces a spatial-for-temporal trade (figure 14). A single tap read over a thirty-frame window recovers the coefficient state as well as eight taps read at one instant, and intermediate combinations of tap count and window meet in between, because the encounter is low-dimensional and a state that filled its ambient dimension would not admit the trade. The eight-tap, thirty-frame recovery reconciles exactly with the static recovery above (0.656), and the delay stride sits at the cache cadence, close to the first mutual-information minimum of the tap signals at 29 frames.

The grid uses a shared target-blind tap array so that the trade is free of any model-conditioned placement; rerunning its eight-tap, thirty-frame cell on the model-conditioned taps of table 8 reproduces that table's recovery exactly (0.656 against 0.659), so the two analyses meet where their sensing coincides. This trade belongs to the static delay-coordinate reconstruction, and it does not transfer to the frozen sequential filter, which needs its eight taps. Rerunning the filter with only its own leading taps, and changing nothing else, drives the analysis lift below the skill of the encounter mean at every gust strength the eight-tap filter tracks: at $|G| = 1$ the single-tap analysis is worse than the eight-tap filter by a paired margin whose interval $[-107.96, -4.78]$ excludes zero, and the loss stays significant at $|G| = 1.5$ and $|G| = 2$. The rank of the innovation, with the process and observation noise held at their calibrated values, is what the filter cannot spare: its sensor budget is a calibration constraint, not a delay-embedding one, which points at the noise model of §5.5 rather than at fewer sensors.

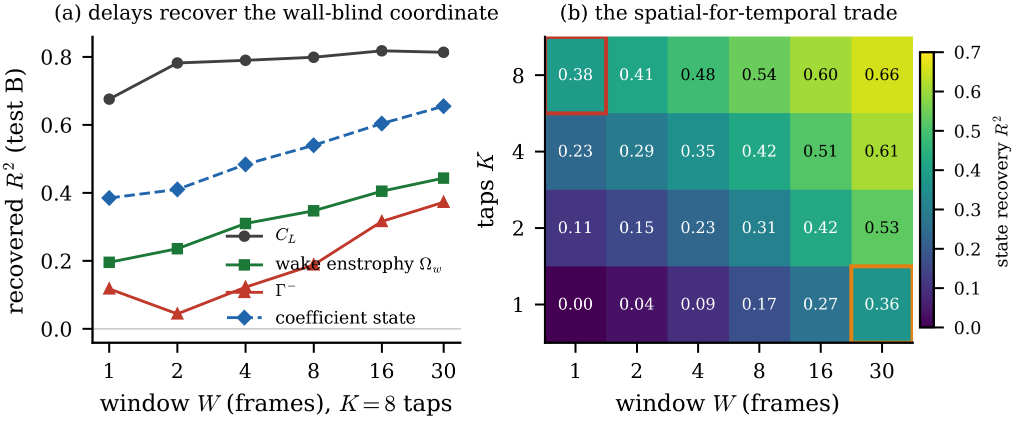


Figure 14. Sensors traded for delays. (a) Recovery of each observable from eight wall-pressure taps against the length of the pressure window (in-distribution test set); the wall-blind wake circulation rises with the window while the force, visible at one instant, does not. (b) Coefficient-state recovery over the tap-count by window-length grid, with the single-tap, thirty-frame cell and the reference eight-tap, single-frame cell outlined: a long window with few taps matches a short window with many.

4.4. *Sequential estimation and operating limits*

4.4.1. *Tracking the encounter from the wall*

We now run the estimator. It is initialised from wall pressure alone before impact and corrected at every frame through the encounter, and it never sees the vorticity field. Figure 15 follows representative held-out encounters, the true lift and wake enstrophy against the open-loop rollout with no correction, the sliding-window pressure-only regression with no dynamics, and the filter analysis. The open-loop rollout tracks the pre-impact baseline and then loses the encounter, as a forecast with no measurement must; the pressure-only regression follows the force while its window is representative and goes stale through the transient; the filter follows the load through impact and into relaxation. This realises, on the wake-supervised coefficient state, the online-estimation pathway the extreme-aerodynamic control literature has named but not closed (Fukami *et al.* 2024; Mousavi & Eldredge 2025; Eldredge & Mousavi 2025): the state a real-time controller would need is recovered from sparse wall pressure and advanced by the learned dynamics, without ever measuring the field.

Two honest boundaries frame what this demonstration does and does not establish, and we state them before the envelope. First, tracking the load inside the training range is not unique to this state. Running the identical frozen filter on every family (table 9), the nonlinear latents all track the lift for $|G|$ between one and two, and the wake-supervised reconstructive control is the best of them at the mildest gusts (0.86 at $|G| = 1$ against 0.74 for the predictive state); only the linear basis fails outright. What no static approach achieves is the strong-gust range: a windowed regression from the same taps straight to the lift, with no latent at all, fades to 0.08 by a gust ratio of three, and even the full wall array recovers only 0.14 at the boundary, so the sequential accumulation of pressure through a learned nonlinear state, rather than any particular training objective, is what carries the load into the strong-gust regime. Where the objectives separate is at that boundary and in the filter's statistical health: the predictive state alone combines the highest boundary closure (0.84 against 0.66 to 0.85 for the alternatives), the lowest divergence rate at every gust strength, and the least-degraded wake readout. Second, no family's filter tracks the wake enstrophy: the analysis wake closure is below the mean predictor for every state, least

Table 9. The frozen filter run identically on every family: median analysis lift closure by gust strength, with the divergence rate at the $|G| = 4$ boundary, and the direct no-latent baseline (windowed regression from the same eight taps straight to the lift) beneath. Load-tracking inside the training range is a property of the nonlinear latents generally, not of any one objective; the families separate at the strongest gusts, in calibration, and in what else the tracked state carries (§4.4.1). The linear basis and the no-latent regression both fail beyond $|G| = 2$.

state	$\lvert G\rvert = 1$	1.5	2	3	4	div. rate at 4
predictive (wake)	0.74	0.83	0.73	0.63	0.84	0.72
AE (wake)	0.86	0.86	0.81	0.49	0.66	0.93
Fukami	0.79	0.87	0.78	0.68	0.76	0.97
Fukami (wake)	−0.39	0.72	0.68	0.77	0.85	0.95
POD	0.53	0.52	−0.33	−0.27	0.25	0.93
direct regression (no latent)	0.41	0.29	0.21	0.08	0.08	–

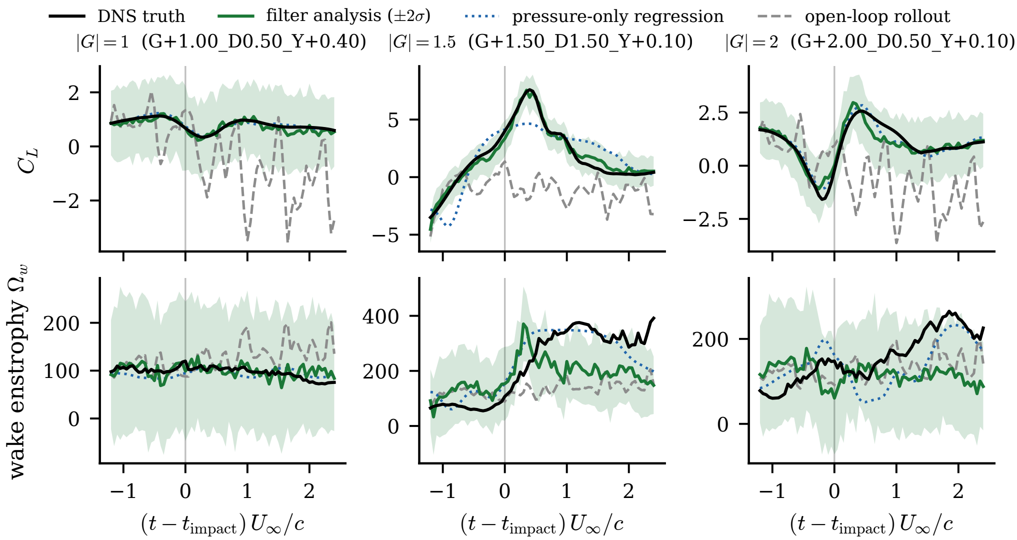


Figure 15. The estimator tracks the encounter from wall pressure alone. Columns are representative held-out encounters at $|G| = 1$, 1.5 and 2; rows are the lift coefficient and the wake enstrophy. Each panel shows the DNS truth, the filter analysis with its $\pm 2\sigma$ ensemble band, the pressure-only regression, and the open-loop rollout; the vertical line marks impact. The filter follows the load through impact and relaxation where the open-loop forecast loses it and the pressure-only regression goes stale.

badly for the predictive one, so the wake content of the encounter is established at the representational tier (§4.1.3) and is not yet recoverable through the assimilation. The filter is a load estimator whose state carries wake information; it is not a wake estimator, and we say so plainly.

How far into the gust the load recovery holds is the operating envelope of §4.4.2.

4.4.2. *The gust-intensity operating envelope*

A reduced state earns its place at deployment only if it can be followed from the sensors that are present, and the wall pressure a small vehicle can carry does not, at a single instant, fix the wake it leaves behind (§4.3). The question is therefore not whether the state is readable but how far into the gust an estimator that sees only the wall can track it. The sequential filter reaches much further than the single-frame inverse. While the encounter stays mild, the static single-frame recovery of the lift holds, with a held-out R^2 of 0.73 at

Table 10. The operating envelope, by gust strength, for the predictive coefficient state under the frozen filter. Median analysis lift closure and divergence rate of the filter against the static single-frame recovery, over all encounters (train, validation, in-distribution test, and the $|G| = 4$ boundary test sets), stratified by $|G|$. The $|G| \leq 0.5$ closure is near zero because the true lift is near-constant there and the R^2 denominator vanishes, not because the filter fails; the per-encounter paired tests on the in-distribution test and boundary test sets are in Appendix A.

$\|G\|$	filter C_L R^2	filter divergence rate	static recovery C_L R^2
≤ 0.5	0.25	0.00	0.34
1	0.74	0.00	0.73
1.5	0.83	0.09	0.65
2	0.73	0.33	0.24
3	0.63	0.68	−0.46
4	0.84	0.72	−0.33

Table 11. What the wall-pressure filter costs in real lift units. Median analysis C_L error (RMSE, dimensionless) and filter divergence rate at the impact window, by gust strength, each family run with its own validation-selected inflation ($\rho = 1.00$ for the predictive filter, 1.05 for Fukami, 1.00 for POD). The variance-normalised R^2 rises with gust amplitude and hides this: at $|G| = 4$ the predictive filter scores $R^2 = 0.84$ while its absolute error is 0.72. Two readings follow. At moderate gusts every filter tracks the load, the reconstruction filter marginally best. At the strongest gusts every filter degrades, with divergence rates of 0.72–0.93; the predictive filter degrades least, in both error and divergence, and the linear filter worst. This is a least-bad regime, not a solved one.

	median C_L RMSE			divergence rate		
Filter	$\|G\|$=1	$\|G\|$=2	$\|G\|$=4	$\|G\|$=1	$\|G\|$=2	$\|G\|$=4
predictive	0.31	0.63	0.72	0.00	0.33	0.72
reconstructive (Fukami)	0.26	0.61	0.79	0.02	0.53	0.95
linear (POD)	0.36	1.14	1.33	0.01	0.49	0.93

$|G| = 1$ and 0.65 at $|G| = 1.5$; as the gust strengthens it collapses, falling below the skill of predicting the encounter mean by a gust ratio of three (−0.46) and remaining there at the $|G| = 4$ boundary (−0.33), because a single pressure frame cannot see the reorganised wake a strong gust deposits. Reading the same taps through the learned dynamics, the filter follows the load further, but the accuracy that matters at deployment is the error in real lift units, not the variance-normalised score, and that error grows with the gust. The median analysis C_L error rises from 0.31 at $|G| = 1$ to 0.68 at a ratio of three and 0.72 at the boundary, even as its R^2 rises to 0.84: the strong gust inflates the lift excursion faster than the error grows, so the normalised score flatters the extreme-gust tracking while the absolute error roughly doubles. The open-loop forecast never tracks the load at any gust strength, so it is the correction step, not the forward model, that recovers the state, and it does so from pressure the visibility analysis found blind to the wake circulation.

Run identically on each family, each under its own validation-tuned inflation, the comparison is a division of degradation rather than a clean win (table 11). At moderate gusts every filter tracks the load, the reconstruction filter marginally best. At the $|G| = 4$ boundary every filter is stressed, diverging on 0.72 to 0.95 of encounters; the predictive filter degrades least, with the lowest median error (0.72 against 0.79 for the reconstruction and 1.33 for the linear basis) and the lowest divergence rate, and the linear filter degrades worst. The predictive state's advantage at the boundary is one of robustness, a least-bad regime rather than a solved one.

What weakens with gust strength is not the tracking but the filter's own account of its uncertainty. The divergence rate climbs from none at $|G| = 1$ to a fraction 0.72 of

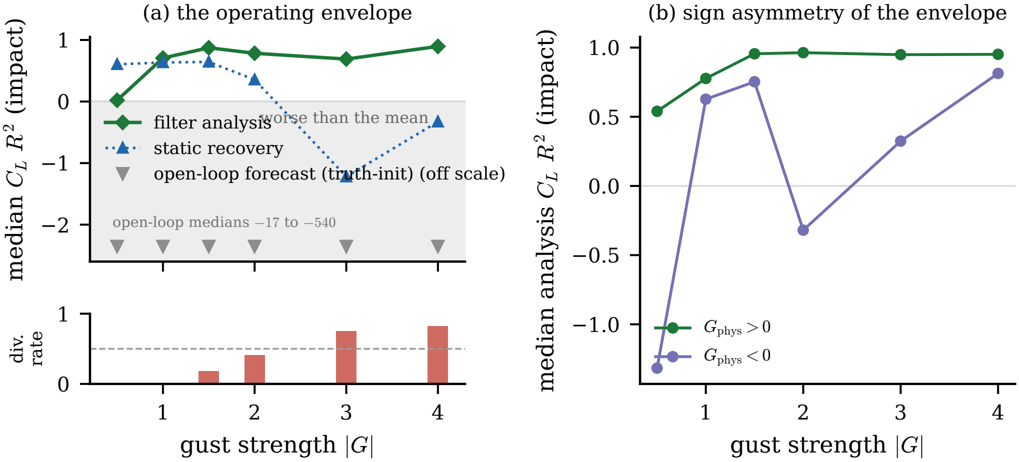


Figure 16. The gust-intensity operating envelope. (a) Median analysis lift closure (R^2, impact window) against gust strength for the frozen filter, the static single-frame recovery, and the truth-initialised open-loop forecast, with the filter divergence rate as the strip beneath; the shaded band marks closure worse than predicting the encounter mean, and the open-loop forecast lies off the scale below it. (b) The same filter closure resolved by the physical sign of the gust, showing the wider usable envelope of the positive gusts.

encounters at the boundary, and the mean normalised innovation squared from near unity in the undisturbed limit (0.6) to nearly twenty at $|G| = 4$ (15.2), crossing the half-divergence mark near a ratio of 3.0 for the compact cores and near 3.0 for the widest. The ensemble collapses while the analysis mean still follows the load, so the filter is under-dispersed rather than mistracking. We read this as the signature of a missing noise model: the innovations are correlated in time rather than white, and the remedy is a calibrated process and observation covariance rather than the covariance inflation we deliberately withheld (§5.5).

The envelope is not symmetric in the sign of the gust, and the asymmetry survives in the linear basis, so it belongs to the flow rather than to the representation (figure 16(b)). The physically positive gusts, which roll up the larger and longer-lived leading-edge vortex (§4.1.4), leave the wider usable envelope. Sampling the $|G| = 4$ boundary at both signs (§2.2) is what lets this asymmetry be measured rather than assumed.

The boundary sits at a gust ratio of three, inside the training range, so it marks a limit of what the mid-plane pressure can observe and the filter assimilate, not a gap in training coverage. Read through delay coordinates the two limits are one. The filter follows the load where the single frame fails because it reconstructs the state from a window of pressure rather than an instant (§3.3), and it loses its calibration where the strengthening gust drives the interaction three-dimensional and the encounter's effective dimension outgrows what a fixed tap count and delay window can embed (§4.3). Only at $|G| = 4$ does the mid-plane representation meet the genuine three-dimensional observability boundary the field itself reaches (§2.1), now seen from the filter. The envelope is therefore two claims with different owners: its extension beyond every static delay-embedded inverse belongs to the sequential estimator on a nonlinear latent, a property the family comparison of §4.4.1 shows is shared; its width at the strongest gusts, its calibration margin, and the wake content of the state being tracked belong to the wake-supervised predictive state. This envelope, not the single-instant readability, is what a wall-estimable reduced state is for.

4.4.3. *The estimator ladder, its calibration and its deployment envelope*

Physical units reorganise the phase story. Read in R^2, the relaxation phase looks like a failure (-1.17 for the filter on the in-distribution test set); read in lift units it is the best phase the estimator has (root-mean-square error 0.18 against 0.30 at impact and 0.29 before it), because the relaxation variance is small and the ratio, not the error, collapses (figure 17a,b). Assimilation earns its keep at impact: the open loop drifts to 1.54 where the filter holds 0.30, a five-fold reduction, with the median peak error at 11.7 per cent of the true peak and peak timing within one frame. Relative accuracy is scale-invariant across the training envelope, near 12 to 14 per cent at every gust intensity for the forecast-derived filter while the linear filter is non-uniform (figure 18); the absolute error grows with the peak it tracks, which is what a scale-invariant estimator must do.

The estimator ladder orders the suite at impact (figure 17c), and its top rung is an honest surprise: the static delay-embedded inverse, no dynamics at all, posts the best in-distribution test impact median (0.83). The filters' case does not rest on the in-distribution median. It rests on the tails, where the static delay-embedded inverse posts the only catastrophic encounter in the ladder; on the extrapolation boundary, where the two-stage filter reaches 0.837 median R^2 on the $|G| = 4$ boundary test encounters with zero catastrophic cases and a positive relaxation (0.120); and on state tracking beyond the single read-out the inverse regresses. The linear filter with encoded observations adds the robustness column: zero divergences under the fixed criterion of §3.3.2 and graceful degradation as the assimilation rate drops, where the transformer-forecast hybrid collapses already at every-second-frame updates (figure in appendix B).

Calibrating the forecast-derived noise did not go the way the calibration hypothesis expected, and the failure is informative. On the held-out training encounters the pooled impact-phase innovation statistic sits below one at every band scale in the pre-registered grid (0.53 at the selected edge $c^* = 1.00$) while the same split's tracking improves monotonically with the band (figure 17e): innovation consistency and tracking optimality point in opposite directions, so the frozen NIS-selected run closes at 0.638 at impact on the in-distribution test set and the consistency route to a larger band is refuted. The mechanism is model error, not miscalibration: inflating the state-dependent noise compensates the forecast median's contraction, buying gain at the price of consistency, and the test-selected variant excluded by the freeze rule is disclosed, with its number, only in appendix C. The member-noise band of the production filter is 0.764 over five seeds. The legitimate gain comes from structure instead of noise: the knob-free two-stage filter of §3.3.3, at the production coverage band, reaches 0.794 at impact on the in-distribution test set (root-mean-square error 0.286) and carries the boundary test set as above, the best protocol-clean filter in the ladder.

When a quarter-convective-time latency is acceptable, the smoother changes the deployment rule. The fixed-lag recursion of §3.3.4 on the linear stack cuts the impact error to 0.286 and the relaxation to 0.149 (filter values 0.375 and 0.141), a full-interval reanalysis at lag five; the ensemble smoother on the forecast-noise filter degrades it (supplementary material). Online estimation therefore takes the nonlinear filter and a $0.25\,t/c$ latency budget takes the closed-form linear smoother.

What the training objective buys survives to the assimilation stage, and own-stack evaluation makes the comparison fair: each family senses at its own taps, encodes its own observations, forecasts with its own operator and decodes through its own decoder. The wake-supervised predictive state halves the reconstructive anchor's assimilated load error in every phase (impact 0.26 against 0.41, relaxation 0.14 against 0.37; the lift-focused predictive state between at 0.31), while decoded-field fidelity is family-insensitive (figure 19a). The sensor axis separates the families further: the predictive state exploits every added tap monotonically where the reconstructive one saturates at $K = 8$, and it degrades more slowly with tap noise, still beating the clean reconstructive stack at twenty per cent noise (figure 19b,c). Under the deployment-realism protocol, streaming multi-encounter assimilation with a pressure-only initialisation and no oracle anywhere, the filter holds 0.789 clean and 0.780, 0.749 and 0.646 at five, ten and twenty per cent tap noise (three member-noise seeds each, figure 20); the four-dimensional lift-critical state of §4.1.2 supports a filter of its own at 0.782 impact median over five seeds.

The closing study runs the whole design axis through one identical own-stack protocol: three families at four dimensions (figure 21). The predictive family is uniform, impact error 0.298 to 0.265 across $d \in \{4, 8, 16, 32\}$ with peak errors of ten to fifteen per cent everywhere; even its $d = 4$ cell (0.298) beats the linear floor's best (0.346 at $d = 32$). The linear basis is stable and a step worse everywhere. The reconstructive baseline is erratic in both directions of the design axis: catastrophic at $d = 4$ and $d = 32$ (1.620 and 2.247, peak errors above one hundred per cent, assimilation at $d = 32$ making the open loop worse), and its celebrated $d = 16$ cell, the best single cell in the grid at 0.180, does not survive seed replication: across three encoder seeds the same cell spans 2.252 impact error on average with individual seeds at 0.18, 0.65 and 5.9, the worst of them failing exactly as the neighbouring dimensions do. The mechanism is verified and lives on the observation side: linear probes on the true reconstructive latents are healthy at every dimension and every seed, while the pressure-to-latent map fails to recover the lift-carrying directions, insensitive to its regularisation, under a protocol that succeeds for the other two families in every cell. Wall-pressure estimatability of the reconstructive geometry is unpredictable across both dimension and training seed; the anchored families are uniformly estimatable, which is the deployment-robustness argument of this paper.

Two loose ends close the section. The under-dispersion of the base filter has two regimes, not one: at impact it was calibration, and the state-dependent noise removes the catastrophic divergences entirely; in relaxation it is model error, the forecast median's contraction, which no noise model fixes and the smoother does. And the wake-enstrophy null of the frozen filter, no family's filter tracks E_w, now has its mechanism: the wall constrains the lift-carrying near-body directions, the ones the Chang decomposition of §3.1.1 weights, while the wake enstrophy lives in shed structure the taps reach only through the delay window, so the null is the sensing geometry, not the state; the delay-for-sensors trade of §4.3 bounds what any wall-limited estimator of this flow can see.

5. Discussion

5.1. *What a wall-estimable state must contain*

The central result is a design rule for a reduced state that a wall-pressure estimator can track. Such a state must carry the wake observables that govern the post-impact transient, must be organised so that a forward model can advance it without leaving its own support, and must be recoverable from the surface pressure that is the only measurement a vehicle carries. These three requirements are supplied by different ingredients, and separating them is the paper's contribution. Observable supervision on the wake supplies the first, the instantaneous readability, and, less obviously, much of the second: it concentrates the encoded variance into a low-rank subspace aligned with the observables, and the forward rollout, trained and living in that subspace, stays in distribution, where an anti-collapse regulariser applied to a bare reconstruction produces a near-isotropic latent with no protected subspace and a rollout that leaks into the directions the training data never constrained (§4.2.4). The mechanism is geometric, and it refines our earlier reading, which credited the anti-collapse term: a reconstruction objective fixes the latent only up to a

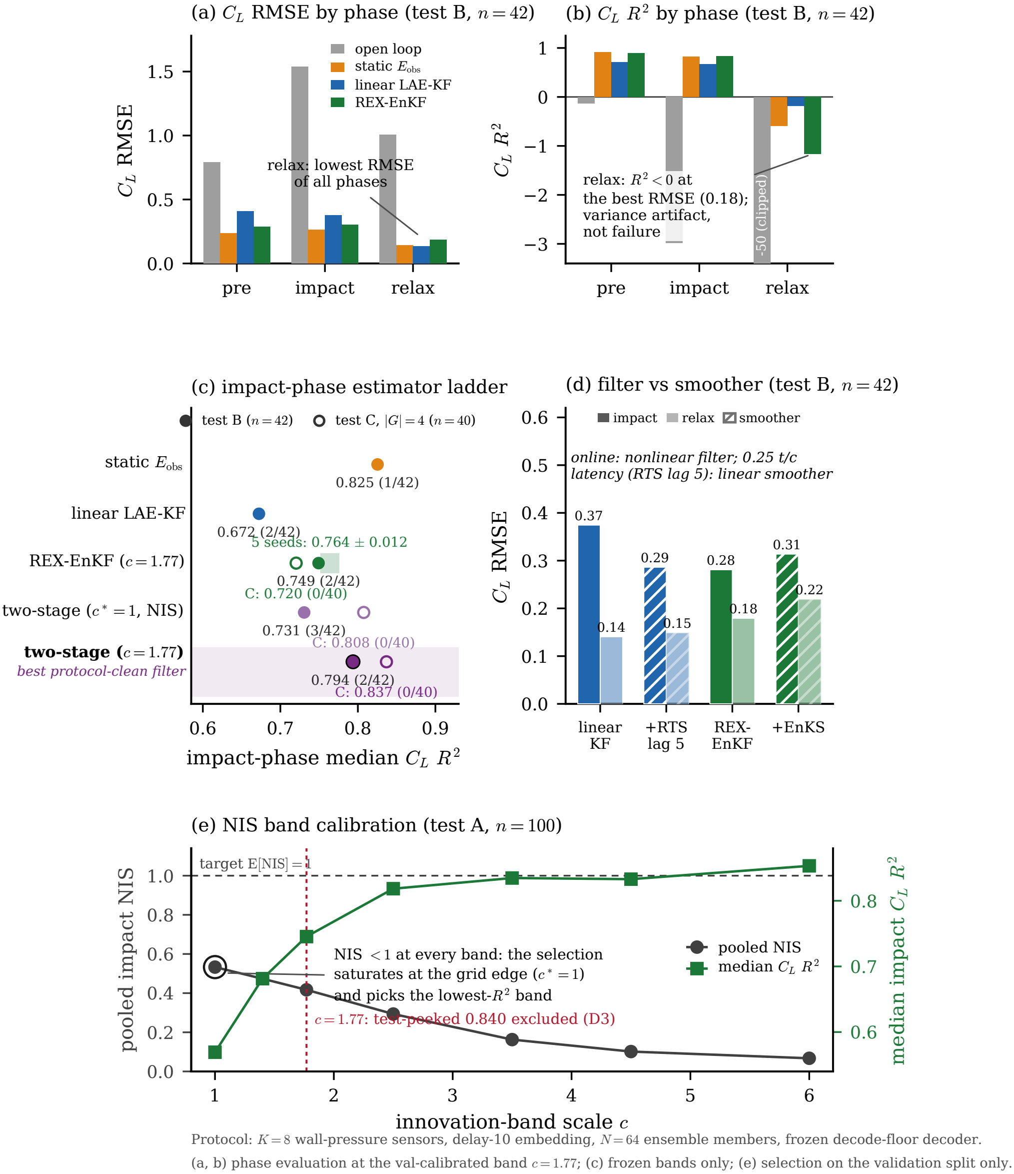


Figure 17. Phase-resolved assimilation on the in-distribution test set (n = 42 encounters, K = 8 taps, delay 10, N = 64 members, every-frame pressure). (a,b) Lift error per phase in physical units and in R^2 at the val-calibrated band: the relaxation R^2 collapse is a variance artifact. (c) The impact-phase estimator ladder with the five-seed member-noise band; open markers are the frozen boundary test values. (d) Filter against fixed-lag smoother. (e) The band-scale calibration on the held-out training encounters: consistency and tracking disagree, and the selection saturates at the grid edge.

diffeomorphism (Kelshaw & Magri 2024), the anti-collapse term on its own drives the latent toward isotropy (Broustail *et al.* 2026), and it is the supervision that pins the variance onto the observable-aligned directions the rollout respects. The multi-step predictive objective supplies the rest of the second requirement, the long-horizon stability that a single-step objective and an objective-free supervised encoder both lack, and, across retrains, the reliability of the forecast itself (§5.5). The third requirement, wall recoverability, is where

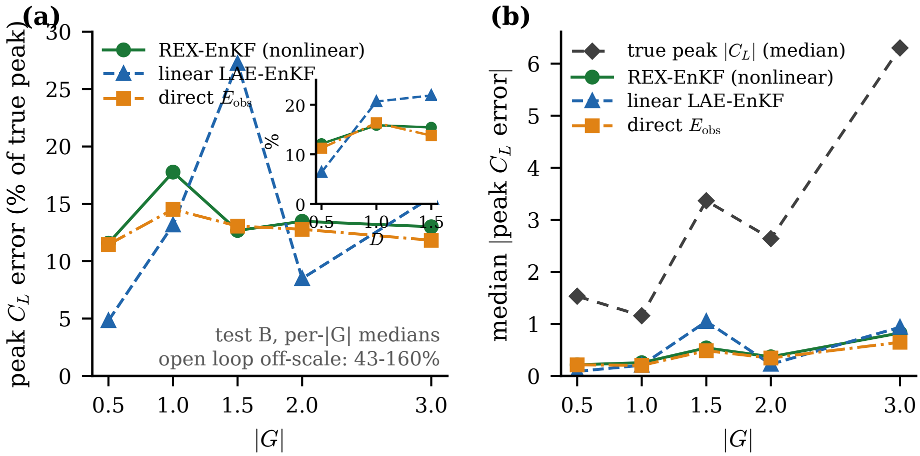


Figure 18. Relative peak error across the gust envelope (in-distribution test set, medians per $|G|$): the forecast-derived filter is scale-invariant near 12 to 14 per cent while the linear filter is non-uniform; the absolute peak error (right) grows with the peak itself.

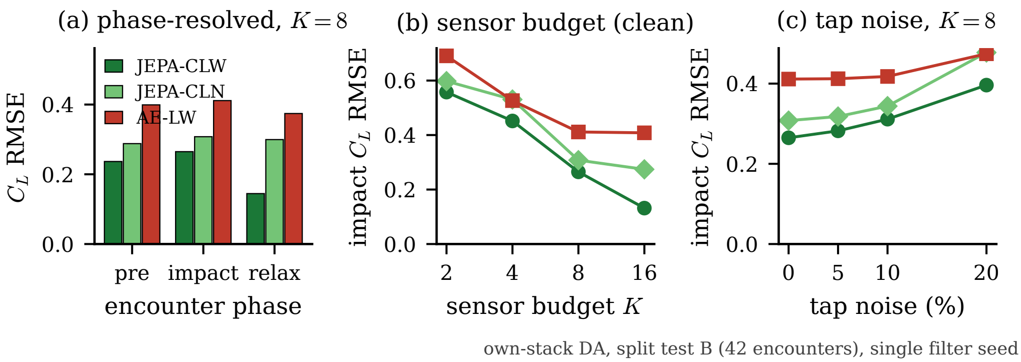


Figure 19. Own-stack family comparison (in-distribution test set, 42 encounters, single filter seed): (a) phase-resolved lift error per family; (b) sensor-budget sweep on each family's own nested placement; (c) tap-noise sweep at $K = 8$.

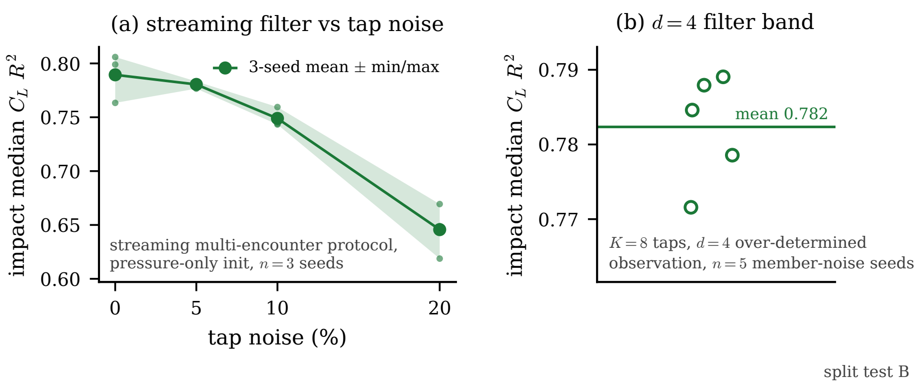


Figure 20. Deployment realism (in-distribution test set): (a) streaming multi-encounter filter against tap noise, three member-noise seeds, protocol-clean band; (b) the $d = 4$ filter band over five member-noise seeds.

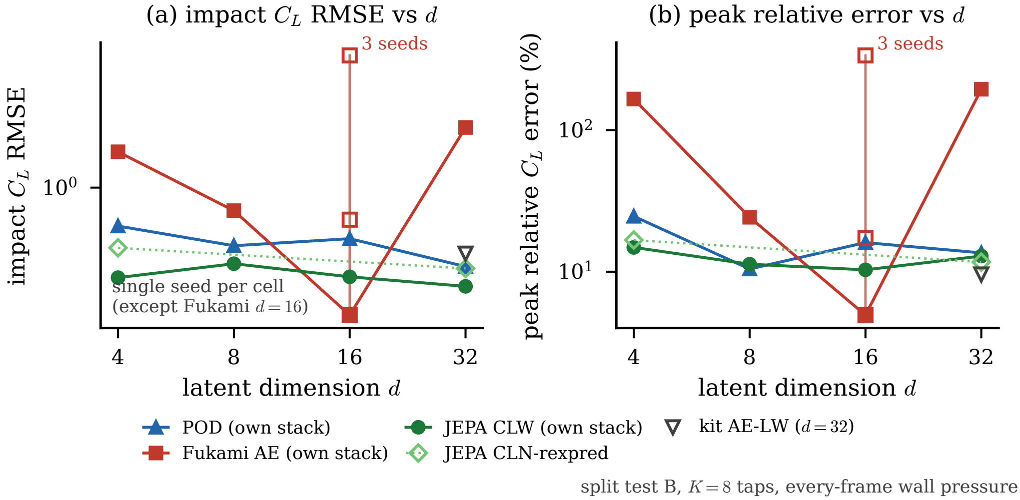


Figure 21. The assimilation-versus-dimension grid (in-distribution test set, own stacks, $K = 8$, every-frame pressure; single seed per cell except the reconstructive $d = 16$ cell, which carries three): impact error and relative peak error against dimension for the three families. The predictive family is uniform, the linear basis stable, the reconstructive baseline erratic in dimension and in seed.

energy and information part company: the reconstruction latent is the most recoverable in raw variance but the least in physics, while the predictive latent carries the most wall-recoverable physics (§4.3), so the coefficient state a controller would want is not the one a reconstruction objective produces.

The wake, and not the forces, is where these requirements separate the families, and the coordinate-level reading of §4.2.2 explains why. The forces are carried redundantly in every family, so an energy or reconstruction objective retains the force signature of an encounter without retaining the distributed combination of coordinates that carries the vorticity redistribution producing it. The gust parameters alone predict the forces well and fail on the wake at the readout horizon, so a state can look accurate on the load while carrying none of the physics the load is built from, which is exactly the failure mode the estimation results expose. The linear basis, for its part, remains genuinely useful: it is competitive on the body forces, amplitude-accurate in reconstruction, and it holds the shedding clock as well as the predictive latent (§4.1.4); the result is not that POD is obsolete but that no linear projection reaches the wake at any dimension we tested.

5.2. *The estimation demonstration and its limits*

The filter turns the design rule into a working estimator: from eight wall-pressure taps and no field measurement it tracks the load through the encounter and extends the usable gust-intensity range beyond every static inverse, realising the online-estimation pathway the extreme-aerodynamic control literature has named (Fukami *et al.* 2024; Mousavi & Eldredge 2025; Eldredge & Mousavi 2025). We are deliberate about its limits, and the first is one the family comparison forces: the in-range load-tracking is a property of sequential estimation on a nonlinear latent generally, not of this state, so the paper does not rest the representational claim on it. What the predictive state contributes to the estimator is the strong-gust boundary, the calibration margin, and a tracked state whose coordinates carry the wake; what no state yet contributes is a filtered wake estimate, since the analysis wake closure is below the mean predictor for every family. Its recoverability advantage over a raw high-dimensional field latent is real but modest (0.120 in held-out R^2 with a case-clustered interval $[0.096, 0.145]$ over the train and in-distribution test recovery rows, the boundary test set untouched; §4.3), so the case for the coefficient state rests on the physics-recovery ordering and the envelope, not on a large recoverability margin. Its point tracking is good but its uncertainty is under-dispersed at the strongest and widest gusts, where the innovations are temporally correlated and the ensemble collapses. The visibility analysis suggests the mechanism: the forward model has large gain along the data-unconstrained near-null directions it never learned to damp, and those inject spread-consuming noise during assimilation. This points to a calibrated process and observation noise model, rather than covariance inflation, as the outstanding requirement, and we report the filter as a feasibility demonstration with a characterised soft spot, not as a deployed estimator.

The delay-embedding view names the deployment knob and its limit. Because the encounter is low-dimensional, the state can be tracked from few taps read over a short window, which is the sensing budget a small vehicle could carry; the trade between tap count and window length is explicit for the static reconstruction (§4.3.1), so a sensor-poor platform can in principle compensate with a longer window within the encounter's own timescale. The frozen filter does not inherit that trade, because its innovation rank, not the embedding, is what binds at reduced tap count, and the two statements together sharpen the same conclusion: the noise model, not the sensor count, is the binding constraint. The limit that no budget escapes is the growth of the encounter's effective dimension with gust strength. Read through delay coordinates the encounter is an intermittently forced low-dimensional system (Brunton *et al.* 2017), the baseline shedding carrying the delay-coordinate dynamics and the vortex impact acting as a forcing whose amplitude is the gust ratio, which is consistent with the envelope being organised by $|G|$: no fixed sensing budget embeds the strongest, most three-dimensional encounters, which is why the envelope closes where it does (§4.4.2).

5.3. *Decodability versus estimability*

A reduced state can be optimised for two deployments that pull apart. A spatially resolved latent decodes the vorticity field more sharply than the coefficient state, but it is of far higher dimension than the wall measurement and so is not recoverable from a sparse set of pressure taps. The coefficient state decodes the field less sharply and is the object a wall-pressure filter can track. We therefore separate the two roles: the coefficient state is the estimable, forward-usable object this paper is about, and the field-resolved latent is a decodability instrument rather than an estimation target. Pooling the encoder onto the coefficient state costs field fidelity across every family, by -0.093 in decode similarity and -0.095 in fitted-operator forecast merit for the predictive latent (supplementary material), and it gains nothing in wall recoverability that the coefficient state does not already have. The state a controller would carry is therefore not the one a field-reconstruction objective produces.

5.4. *Choosing the estimator*

The ladder of §4.4.3 supports explicit selection guidance, in the spirit of the method-choice discussions of the assimilation literature (Gupta, Chen and Wan, J. Fluid Mech. 1036, A24, 2026). When only the in-distribution median matters and latency is irrelevant, the static delay-embedded inverse is the strongest and simplest tool, and no dynamics are needed. The moment the tails, the extrapolation boundary or continuous state tracking matter, the filters take over: online, the two-stage filter is the working choice, the forecast-derived noise carrying the impact and the tap-space observation carrying the relaxation; with a quarter-convective-time latency budget, the closed-form fixed-lag smoother on the linear stack is both the cheapest and the best reanalysis. Sparse or intermittent pressure favours the linear filter with encoded observations, the only member with zero divergences and graceful degradation under subsampled updates; every-frame pressure is load-bearing for the nonlinear members. Noise calibration should be spent on structure before scale: the state-dependent band fixed the impact-side divergences outright, while inflating its scale toward better tracking is model-error compensation that innovation consistency rejects, so a larger band should be read as a diagnostic of forecast contraction, not as a calibration target. Across model families the guidance is one sentence: anchored latent geometries, predictive or linear, can be selected on dimension and budget alone, while unanchored reconstructive geometries require a per-configuration estimatability check that their probes do not predict.

5.5. *Limitations and outlook*

Five limitations bound the claims. First, the attribution of the instantaneous results is unglamorous: the wake readability is supplied by the observable supervision and its tie between the predictive and reconstructive wake-headed states is seed-robust (0.727 ± 0.021 against 0.760 ± 0.019). The shared-operator forecast merit of §4.1.3 shows the same picture dynamically: over three operator seeds the three wake-headed states are statistically indistinguishable, with overlapping case-clustered intervals. An earlier seed study of the fitted merit, run under the residual U-Net operator as a stability check, cautions the same way: the supervised control is seed-fragile (spread 0.163, with one healthy retrain in three collapsing), the predictive latent is tighter but not the sharpest (± 0.098), and the family means overlap. No fitted-operator margin between the wake-headed states is therefore decisive on its own. The defensible dynamical claim rests instead on the co-trained model and its rollout: the wake-supervised predictive state's own predictor forecasts better than the shared operator extracts from any latent at short horizon (0.755 at horizon eight, §4.1.3), and the multi-step objective confers long-horizon stability and lower drift, not a larger instantaneous readability margin. Second, the wall-normal offset Y remains the least resolved release parameter because the training mass concentrates near $Y \approx 0$; the additional cases make it weakly readable where it was unreadable before (§4.1.4), and we note this without leaning on it. Third, the $|G| = 4$ extrapolation degrades for a physical reason: the encoder observes a single mid-plane vorticity slice whose information no longer determines the three-dimensional dynamics there (§2.1), so the boundary test set marks the observability boundary of that representation, not a parametric-generalisation claim, and the filter meets the same boundary from the sensing side (§4.4.2). Fourth, the visualisation decoder is a diagnostic on the frozen encoder, never part of the predictive loss, and the pooled coefficient state trades decode sharpness for estimability by construction (§5.3). Fifth, and foremost for what comes next, the filter's calibration is the priority open problem: its uncertainty is under-dispersed at the strongest gusts, and the natural next step is a learned or physically motivated process and observation covariance, ahead of any closed-loop use.

These limitations come into sharpest focus against the use the architecture is meant for. The predictor is, by construction, a latent dynamics function of the kind an estimator or controller requires, the latent is orders of magnitude cheaper to advance than the field, and the filter demonstrates the estimation half of that loop from the wall alone. Closed-loop actuation is outside the present evidence: the predictor is trained to forecast observed encounters within the training envelope, and predicting how an encounter would differ under intervention is a task its training does not address. The contribution is narrow by design, a controlled comparison and a frozen-filter characterisation on a single parametric flow, and what it establishes is bounded accordingly: a wall-estimable, forward-usable coefficient state, the ingredients that produce it, and the gust-intensity envelope over which a fixed sensing budget can follow it. The natural next steps are the noise model above and the same matched-estimator protocol on other parametric flows, to test how general the estimability ordering reported here actually is.

6. Concluding remarks

We asked whether a reduced state of an extreme vortex-gust encounter can be constructed to carry the wake dynamics, advanced in time, and recovered from wall pressure alone. On construction: under the predictive objective considered, a scalar lift target is necessary for a non-collapsed latent representation, a near-body force-element target sharpens the peak loads on top of it (peak-region $R^2 = 0.86$ over three seeds), and a wake target is required for the state to carry the wake observables. The collapse result concerns latent health and the peak-load result concerns a paired increment over an already lift-supervised baseline; the two rest on different comparisons.

On compression, the statement is two-tier: load information is approximately unchanged over the tested dimensions under a nonlinear probe and is linearly accessible at four coefficients, whereas wake-field reconstruction requires at least sixteen. On forecasting, a direct multi-horizon forecaster remains usable over forty steps where autoregressive rollout degrades through error accumulation, and supplying the true gust parameters provides no benefit, so the deployable route to the encounter is a model corrected by pressure rather than the release parameters.

On estimation, a two-stage ensemble filter sensing eight wall-pressure taps tracks the lift through the encounter, phase-resolved and in physical units: impact-phase $R^2 = 0.794$ at a root-mean-square error of 0.286 lift units on the in-distribution test set, and $R^2 = 0.837$ at the $|G| = 4$ boundary. Across latent dimension and training seed the predictive states remain uniformly estimable from the wall, whereas the reconstructive geometry does not, for a verified observation-side reason; this robustness, rather than any single configuration, is the deployment argument.

Three results bound the claims. A static delay-embedded inverse matches the filter's in-distribution impact median, and the filter's advantage lies in the tails, at the boundary set and in continuous state tracking. The wake enstrophy is not recovered by the tap counts, delay windows and estimators considered here, a limit we attribute to the sensing geometry rather than to the state. And the mid-plane representation shows an indication of a three-dimensional observability limitation at the strongest gusts (four cases), where the filter's uncertainty consistency degrades even as the median tracking holds. The natural next steps are a calibrated process and observation noise model for the filter, and the same matched-estimator protocol on other parametric flows, to test how general the estimability ordering reported here is.

Funding. *[Authors to insert funding sources and grant numbers.]*

Declaration of interests. The authors report no conflict of interest.

Data availability statement. The reduced-order-model and evaluation code, the configuration files, the v2.1 data-split manifest, and the analysis outputs that regenerate the tables and figures are deposited at https://doi.org/10.xxxx/zenodo.PLACEHOLDER under the MIT License (DOI to be finalised on acceptance). The processed per-encounter cache, or a representative subset sufficient to rerun the heavier steps, is available from the corresponding author. The raw direct numerical simulations were computed with the SOD2D solver (Gasparino *et al.* 2024) and are owned by the simulation collaborators.

Author contributions. *[Authors to insert contributions, for example in CRediT format.]*

Use of generative-AI tools. Generative-AI assistance (Anthropic Claude, accessed through the Claude Code command-line interface) was used during 2026 to help with data analysis, figure generation, and manuscript drafting. All numerical results were computed by the authors' own code and verified against the underlying data, and the authors take full responsibility for the content of the article.

Appendix A. Architecture, anti-collapse regularisation, and uncertainty quantification

Architecture detail

The encoder is a hybrid convolutional stem followed by a small vision transformer. Three downsampling convolutional stages map the single-channel 192×96 vorticity field to a 24×12 feature map at 256 channels, that is 288 image tokens, which a six-layer pre-norm transformer of width 256 with eight heads processes; the latent is read from a prepended class token through a one-layer projection whose output passes a BatchNorm, the layer the characteristic-function regulariser below requires. The encoder is unconditional and carries no information about the gust parameters. The training predictor advances the pooled 32-vector directly: its function class is a six-layer autoregressive transformer of width 384 with sixteen heads, rotary temporal positions and a causal mask, mapping a short history of 32-vectors to the next 32-vector, with no spatial map or tiling anywhere in the loop. Training minimises an open-loop rollout loss over eight steps, with no teacher forcing, against online encoder targets that are detached (no exponential-moving-average target network), plus the regulariser below, with AdamW and bf16 mixed precision. The predictor advances the latent from its own history alone, with no gust input, so the gust parameters $\mathbf{c} = (G, D, Y)$ enter nowhere in the model. Two downstream forecast operators are fitted afterwards on each frozen latent, identically across families. Every family produces a pooled coefficient vector at the same dimension, and two matched forecast operators apply to all of them: an autoregressive transformer on the vector sequence, and a residual U-Net that tiles the vector to a grid and forecasts it convolutionally. The forecast merit of table 6 reports each latent under the operator that is numerically stable on it. The transformer is stable across the pooled states of the controlled matrix, forecasting the predictive, supervised, and reconstructive-autoencoder latents comparably and placing the wake-supervised predictive state at the head of that block; but refitted with the same recipe and budget on the reference latents (the β-VAE and the two Fukami autoencoders, which are plain convolutional autoencoders without skip connections) it diverges to large negative merit. The residual U-Net is stable on every latent, so it is the operator reported for the reference latents; imposing either operator on all families would either penalise the pooled states off-class or break on the references. The ensemble filter of §3.3 uses the same matched transformer on the wake-supervised predictive state's own pooled trajectories, where it is stable, and the wake-supervised predictive state's own co-trained predictor, the as-built reduced-order model, is reported alongside. The narrow spread of the pooled block under the transformer, and the substantial seed variation of the merit, are why we read the fitted-merit ordering as a level with a small predictive edge rather than a decisive separation.

Anti-collapse regularisation

Because the encoder and predictor are trained end to end on latent-space prediction with no decoder, the latent is held away from a collapsed solution by an explicit regulariser rather than by a reconstruction term. We use SIGReg, the characteristic-function (Epps–Pulley) regulariser of the from-pixels joint-embedding line (Maes *et al.* 2026; Balestriero & LeCun 2025), with M random one-dimensional projections of the latent and a fixed weight $\lambda_S = 0.02$ in the loss, the value used in every run reported here. The same regulariser has since been adopted independently for biosignal time series, where it is likewise reported to drive the embedding toward isotropy so that a structuring objective is needed alongside it (Broustail *et al.* 2026), consistent with the supervision-anisotropy mechanism of §4.2.4. As a safeguard the participation ratio of the latent is monitored every thousand iterations, with an automatic fall-back to a variance-covariance objective (Bardes *et al.* 2022) if it indicates collapse while the parameter probe is still uninformative; on the runs reported here the participation ratio stays healthy and the fall-back does not trigger.

Uncertainty quantification

Every held-out coefficient of determination and mean absolute error in the paper carries three independent uncertainty signals, following the reporting protocol fixed with the split. The first is a bootstrap confidence interval from 2000 resamples over the held-out test encounters, which is the dominant source of scatter at the present sample size ($n = 42$ for the in-distribution test set, $n = 40$ for the boundary test set) and is wide for the individual wake observables, as noted in §4.1.3. The second is the variance across three encoder retrains from independent random seeds, reported as the standard deviation in the objective-times-architecture controls of §4.2.4. The third is a five-fold cross-validation over cases on the read-out (probe) step, which guards against a probe that overfits the particular held-out split. Together these approximate what a full five-fold case-level

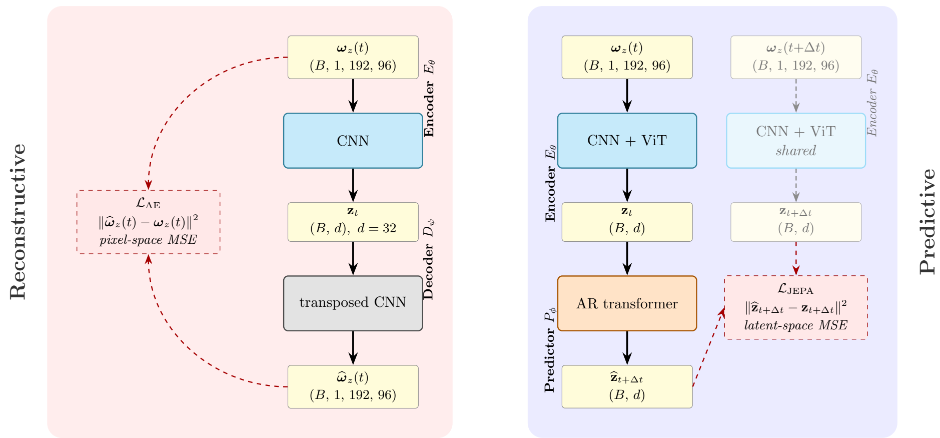


Figure 22. Predictive versus reconstructive training. Left: the reconstructive autoencoder encodes the field at t, decodes it, and incurs a field-space error. Right: the predictive model encodes the fields at t and $t + \Delta t$ with shared weights and incurs a latent-space error between the predicted and observed target embeddings, with no field-space loss and no decoder during training. This is the contrast the matched-predictor protocol of §3.2 evaluates; the visualisation decoder is trained only afterwards, on the frozen encoder.

cross-validation of the entire pipeline would buy, at a fraction of the training cost; a single split with bootstrap intervals is the community standard for this class of study.

Training-set fit

The predictive latent attains the highest in-distribution training fit of the forecast rollout among the families, but this is an in-distribution number, not the held-out result: the family ordering is established by the held-out representational closure of table 6, and the held-out forecast mean is markedly lower than the training fit, as it must be. We therefore do not tabulate the training-set fit, which would invite reading an in-distribution number as a result.

Paired tests on the estimation endpoints

Table 12 reports the pre-registered per-encounter paired family behind the tracking and envelope claims of §4.4.1 and §4.4.2. Pairing cancels the encounter-to-encounter difficulty that widens marginal intervals, and clustering by case respects the shared shedding history of a case's encounters.

Topology of the encoded encounter

A coordinate-free check on the geometric mechanism of §4.2.4: Vietoris–Rips persistence of the encoded trajectories, per family, with a generator declared significant when its persistence exceeds a noise floor set at a fraction of the cloud diameter, after per-family standardisation so that coordinate scale cannot carry the result. On the gusted held-out encounters the supervised states keep a single clean H_1 cycle in about half the encounters (0.64 for the predictive and 0.52 for the objective-free supervised state) where the anti-collapse reconstruction fragments (0.10); on the single-period no-gust control the same separation holds (0.62 against 0.00), and it survives whitening, so the fragmentation is a property of the reconstructive encoding rather than of any coordinate scale. The linear basis keeps the cleanest no-gust loop of all (0.62), consistent with a linear projection of a smooth field; the claim is that the anti-collapse reconstruction fragments, not that only the predictive latent carries a loop.

Preprocessing robustness

The wake enstrophy squares the vorticity, so the per-encounter high-percentile amplitude clip of the ω_z preprocessing could in principle shape the primary endpoint. We test this on the frozen encoders, with no retraining, by recomputing the readout-frame wake-enstrophy target under three training-only clip treatments of the raw field and re-fitting the frozen-latent-to-wake ridge probe per family: no amplitude clip, the production per-encounter $p_{99.99}$ clip, and a single leak-free $p_{99.99}$ threshold pooled over the training encounters. The

Table 12. Paired per-encounter tests for the estimation endpoints, on the frozen filter records. The family was pre-registered before any statistic was computed: the primary family F1 compares the filter analysis against the static single-frame recovery and F2 against the field-free open-loop forecast, on the endpoints $\{C_L, E_w\}$ in the impact and relaxation windows, one-sided with the filter pre-registered as better; $\varDelta R^2$ is the median per-case mean improvement, the p is a case-level Wilcoxon signed-rank, and Holm corrects within each family. On the pooled in-distribution test encounters the filter beats the open-loop forecast decisively on the load but not the static recovery, which is accurate at the moderate gust strengths the in-distribution test set samples; the filter's advantage over the static delay-embedded inverse concentrates at strong gusts, where the boundary test annex (characterisation only, $n = 8$ cases) is decisive on the load in both windows. The wake enstrophy through the filter does not beat the static recovery at the boundary, an honest null.

Family	Endpoint	Window	median $\varDelta R^2$	Wilcoxon p	Holm p
F1 (vs static)	C_L	impact	0.16	0.161	0.483
F1 (vs static)	C_L	relax.	0.53	0.116	0.465
F1 (vs static)	E_w	impact	−25.15	0.986	1.000
F1 (vs static)	E_w	relax.	−2.47	0.839	1.000
F2 (vs open loop)	C_L	impact	9.43	0.001	**0.003**
F2 (vs open loop)	C_L	relax.	44.84	0.001	**0.003**
F2 (vs open loop)	E_w	impact	17.39	0.246	0.246
boundary test annex	C_L	impact	1.11	0.0078	–
boundary test annex	C_L	relax.	6.41	0.0039	–

Table 13. Preprocessing robustness of the representational wake closure, established on the previous-generation spatial encoders ($d = 64$, the 85-case v2.1 split) and retained as evidence that the amplitude clip does not shape the endpoint; the pipeline is unchanged in the present dataset revision, and the probe protocol is identical. Held-out in-distribution test wake-enstrophy R^2 from a frozen-latent ridge probe (no retraining), recomputed under three training-only amplitude-clip treatments of the raw ω_z field. The predictive latent leads the wake under every treatment; the advantage keeps its sign and approximate magnitude.

Family	no clip	per-encounter	training-global
predictive	0.75	0.80	0.84
reconstructive (Fukami)	0.12	0.10	0.08
linear (POD)	-0.27	-0.05	0.00

predictive latent leads the wake on the in-distribution test set under all three (table 13), with the largest shift in the predictive-minus-best-baseline advantage relative to the production clip being 0.07 in R^2. The leak-free training-global clip gives the largest predictive advantage, so the per-encounter clip is not the source of the wake result.

Appendix B. Sparse wall-pressure state estimation

At deployment the vorticity field is not available; only the wall pressure is. This appendix records the sensing conventions behind §4.3 and §4.4.1 and the placement-robustness checks that the main text summarises. The input throughout is the span-averaged pressure on the 192 surface taps; the recovery estimator is selected per family by encounter-grouped cross-validation among a kernel-ridge regressor on the flattened window, a multilayer perceptron, and a recurrent network over the window sequence, and the recovered state is read by the same frozen probes as everywhere else.

Placement, and why it does not drive the ordering

Two placements are used, deliberately. The filter and the recovery of table 8 sense at each family's own model-conditioned taps, the greedy selection that best exposes that family's latent, because the estimator comparison should give every family its best sensing. The delay-trade grid of §4.3.1 instead uses a single target-blind array, the QR-pivot optimum of the wall-pressure modes (Manohar *et al.* 2018), nested so that every tap count is a prefix of the next, because a sensor-budget claim must not depend on a placement tuned to any model. The two

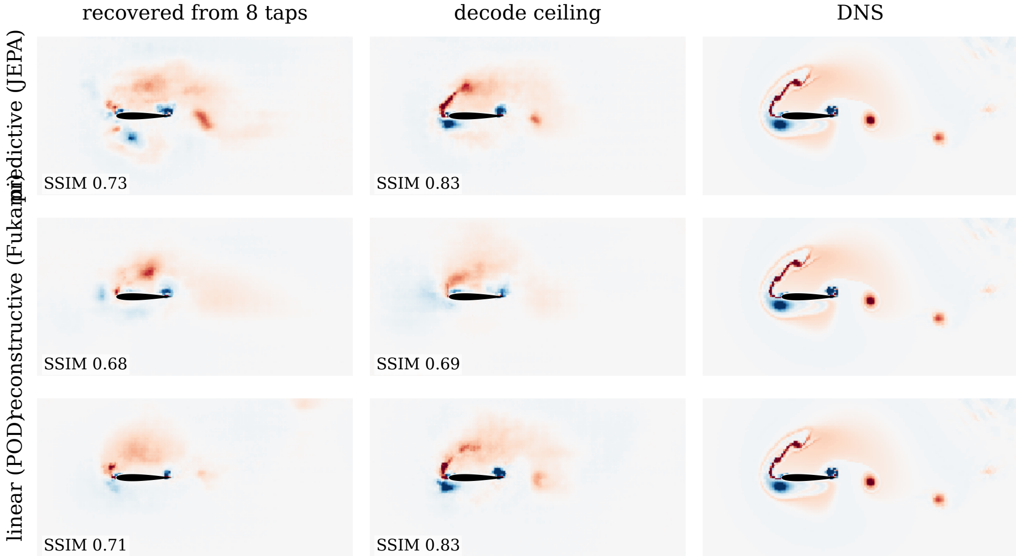


Figure 23. Flow recovered from eight wall-pressure taps on the representative held-out encounter of figure 15, at the impact-side frame, one row per family. Columns: the field decoded from the pressure-recovered coefficient state, the decode ceiling (the same decoder applied to the encoded state, no pressure step), and the simulation; per-panel structural similarity annotated. The comparison is within each row, since the decoders' ceilings differ (§5.3); the ceiling-to-recovered gap is the cost of the pressure step.

meet where their sensing coincides: the grid's eight-tap, thirty-frame cell rerun on the model-conditioned taps reproduces the table's recovery exactly (0.656). The energy-versus-information split of §4.3 is also placement-robust: under the shared target-blind array with the kernel-ridge estimator alone, the reconstructive state remains the most recoverable in variance (0.775 against 0.667 for the predictive and 0.500 for the linear basis) and the predictive state retains the most recoverable physics (0.523 against 0.363 and 0.391), so neither ordering is an artefact of the model-conditioned placement.

What the window recovers that the instant cannot

At a single frame the wall is blind to the wake-circulation direction of the predictive state (§4.3); over the thirty-frame window the blind coordinate becomes partially recoverable (0.33 held-out, from near zero at one frame), and the wake enstrophy read through the recovered state reaches 0.40. The window, not the tap count, is what carries this: the same quantities barely move as taps are added at a fixed single frame. This is the delay-coordinate content of the sensing result, quantified in the grid of §4.3.1; it is a statement about the static reconstruction, and the filter inherits the sequential version of it through its assimilation window rather than through any explicit delay embedding.

The recovered field

Decoding the pressure-recovered state through each family's diagnostic decoder makes the recovery visible in physical space (figure 23). The comparison within each row, recovered field against that family's own decode ceiling, isolates the pressure step from the decoder's reconstruction limit: the loss from ceiling to recovery is the part attributable to the eight-tap window, and the panels show it concentrating in the wake structures rather than in the near-body field, consistent with the wall reading the force-carrying directions best (§4.3).

The out-of-distribution boundary from the sensing side

At the $|G| = 4$ boundary every static approach fails on the load, the via-latent recovery at -0.33 and the direct no-latent regression at 0.08 from the same taps (0.14 with the full array), while the filters still track it (table 9). The boundary is the same three-dimensional observability limit the field reaches (§2.1): the mid-plane state no longer determines the dynamics there, so a static delay-embedded inverse has nothing to invert, and only the accumulated pressure history carries the load through.

Delay-coordinate knobs

The window sweeps of §4.3.1 keep the delay stride at the cache cadence ($\Delta t = 0.05\,c/u_\infty$), and this choice is a convention rather than a tuning: the first minimum of the time-lagged mutual information of the tap signals (Fraser & Swinney 1986) sits near 29 frames, about one shedding period, so the thirty-frame window spans roughly one decorrelation time of the wall pressure at the cadence used. The window length itself is selected by the measured recovery trade rather than by a false-nearest-neighbours criterion (Kennel *et al.* 1992), which we regard as a cross-check and not a design input: the delay-embedding guarantees are generic, wall pressure is a physical and non-generic observable strongly coupled to the near-body vorticity, and the operative evidence is the measured recovery, with the theorem as the organising principle rather than a proof that a given tap count suffices.

Appendix C. Calibration disclosure

The forecast-derived filter's band scale carries a full audit trail. The production value $c = 1.77$ is the coverage calibration of §3.2.1, selected on validation before any test encounter was seen. During exploration a band of $c = 4$ was evaluated against the in-distribution test set and reached 0.840 impact median there; because that value was selected with the test split in view, it is excluded from every table and headline of this paper and quoted only here. The pre-registered consistency calibration (grid and rule committed before any value was seen) subsequently showed that the innovation statistic rejects large bands on held-out training encounters even though their tracking is better (§4.4.3): the large-band gain is real and reproducible on encounters never used for selection, but it is model-error compensation, not calibration, and the paper's protocol-clean route to comparable performance is the two-stage filter instead. The two-stage filter run at the coverage band was itself declared in the pre-registration record after the consistency-selected run was seen and before any run at the declared configuration existed; both frozen runs are reported in §4.4.3 regardless of outcome, and the declaration order ships with the data record.

References

Assran, M., Duval, Q., Misra, I., Bojanowski, P., Vincent, P., Rabbat, M., LeCun, Y. & Ballas, N. 2023 Self-supervised learning from images with a joint-embedding predictive architecture. In *CVPR*.

Assran, M. & others 2025 V-JEPA 2: self-supervised video models. *arXiv preprint arXiv:2506.09985* .

Auer, Andreas, Podest, Patrick, Klotz, Daniel, Böck, Sebastian, Klambauer, Günter & Hochreiter, Sepp 2025 TiRex: zero-shot forecasting across long and short horizons with enhanced in-context learning. *arXiv preprint arXiv:2505.23719* .

Balestriero, R. & LeCun, Y. 2025 LeJEPA: provable and scalable self-supervised learning without the heuristics. *arXiv preprint arXiv:2511.08544* .

Bardes, A., Ponce, J. & LeCun, Y. 2022 VICReg: variance-invariance-covariance regularization for self-supervised learning. In *ICLR*.

Beck, Maximilian, Pöppel, Korbinian, Spanring, Markus, Auer, Andreas, Prudnikova, Oleksandra, Kopp, Michael, Klambauer, Günter, Brandstetter, Johannes & Hochreiter, Sepp 2024 xLSTM: extended long short-term memory. *arXiv preprint arXiv:2405.04517* .

Broustail, Danaé, Tegon, Anna, Ingolfsson, Thorir Mar, Li, Yawei & Benini, Luca 2026 LuMamba: latent unified Mamba for electrode topology-invariant and efficient EEG modeling. *arXiv preprint* **arXiv:2603.19100**.

Bruna, Joan & Mallat, Stéphane 2013 Invariant scattering convolution networks. *IEEE Transactions on Pattern Analysis and Machine Intelligence* **35** (8), 1872–1886.

Brunton, Steven L., Brunton, Bingni W., Proctor, Joshua L., Kaiser, Eurika & Kutz, J. Nathan 2017 Chaos as an intermittently forced linear system. *Nature Communications* **8**, 19.

Chen, Dashuai, Liang, Aoming, Sun, Boai, Rival, David E. & Fan, Dixia 2026 Bridging spatial and temporal surface pressure dynamics for gust aerodynamic modeling. *Communications Engineering* .

Drmač, Zlatko & Gugercin, Serkan 2016 A new selection operator for the discrete empirical interpolation method: improved a priori error bound and extensions. *SIAM Journal on Scientific Computing* **38** (2), A631–A648.

Durall, Ricard, Keuper, Margret & Keuper, Janis 2020 Watch your up-convolution: CNN based generative deep neural networks are failing to reproduce spectral distributions. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. ArXiv:2003.01826.

Eldredge, Jeff D. & Mousavi, Hanieh 2025 A practical guide to estimation and uncertainty quantification of aerodynamic flows. *arXiv preprint* **arXiv:2502.20280**.

Evensen, Geir 2003 The ensemble Kalman filter: theoretical formulation and practical implementation. *Ocean Dynamics* **53** (4), 343–367.

Fraser, Andrew M. & Swinney, Harry L. 1986 Independent coordinates for strange attractors from mutual information. *Physical Review A* **33** (2), 1134–1140.

Fukagata, Koji & Fukami, Kai 2025 Compressing fluid flows with nonlinear machine learning: mode decomposition, latent modeling, and flow control. *Fluid Dynamics Research* **57** (4), 041401.

Fukami, Kai, Nakao, Hiroya & Taira, Kunihiko 2024 Data-driven transient lift attenuation for extreme vortex gust–airfoil interactions. *J. Fluid Mech.* **992**, A17.

Fukami, Kai, Smith, Luke & Taira, Kunihiko 2025 Extreme vortex-gust airfoil interactions at Reynolds number 5000. *Phys. Rev. Fluids* **10**, 084703.

Fukami, Kai & Taira, Kunihiko 2023 Grasping extreme aerodynamics on a low-dimensional manifold. *Nat. Commun.* **14**, 6480.

Fukami, Kai & Taira, Kunihiko 2025 Observable-augmented manifold learning for multi-source turbulent flow data. *J. Fluid Mech.* **1010**, R4.

Gasparino, L., Spiga, F. & Lehmkuhl, O. 2024 SOD2D: A GPU-enabled spectral finite elements method for compressible scale-resolving simulations. *Computer Physics Communications* **297**, 109067.

Haughn, Kevin P. T., Harvey, Christina & Inman, Daniel J. 2024 Deep learning reduces sensor requirements for gust rejection on a small uncrewed aerial vehicle morphing wing. *Communications Engineering* **3** (1), 53.

Jones, A. R., Cetiner, O. & Smith, M. J. 2022 Physics and modeling of large flow disturbances: discrete gust encounters for modern air vehicles. *Annu. Rev. Fluid Mech.* **54**, 469–493.

Kelshaw, Daniel & Magri, Luca 2024 Proper latent decomposition. *arXiv preprint arXiv:2412.00785* .

Kennel, Matthew B., Brown, Reggie & Abarbanel, Henry D. I. 1992 Determining embedding dimension for phase-space reconstruction using a geometrical construction. *Physical Review A* **45** (6), 3403–3411.

Lazebnik, Svetlana, Schmid, Cordelia & Ponce, Jean 2006 Beyond bags of features: Spatial pyramid matching for recognizing natural scene categories. In *2006 IEEE computer society conference on computer vision and pattern recognition (CVPR'06)*, , vol. 2, pp. 2169–2178. IEEE.

LeCun, Yann 2022 A path towards autonomous machine intelligence. *Tech. Rep.*. OpenReview, version 0.9.2.

Loiseau, Jean-Christophe, Noack, Bernd R. & Brunton, Steven L. 2018 Sparse reduced-order modelling: sensor-based dynamics to full-state estimation. *Journal of Fluid Mechanics* **844**, 459–490.

Maes, P.-A., Garrido, Q., Assran, M., Bardes, A., Rabbat, M., LeCun, Y. & Ballas, N. 2026 LeWM: latent world models for self-supervised video prediction. *arXiv preprint arXiv:2603.19312* .

Manohar, Krithika, Brunton, Bingni W., Kutz, J. Nathan & Brunton, Steven L. 2018 Data-driven sparse sensor placement for reconstruction: demonstrating the benefits of exploiting known patterns. *IEEE Control Systems Magazine* **38** (3), 63–86.

Mohamed, Abdulghani, Marino, Matthew, Watkins, Simon, Jaworski, Justin & Jones, Anya 2023 Gusts encountered by flying vehicles in proximity to buildings. *Drones* **7** (1), 22.

Mousavi, Hanieh & Eldredge, Jeff D. 2025 Low-order flow reconstruction and uncertainty quantification in disturbed aerodynamics using sparse pressure measurements. *Journal of Fluid Mechanics* **1013**, A41.

Odaka, Hiroto, Lopez-Doriga, Barbara & Taira, Kunihiko 2026 Vortical similarities across laminar and turbulent extreme gust encounters. *J. Fluid Mech.* **1031**, R3.

Podest, Patrick, Pichler, Marco, Bürger, Elias, Zólyomi, Levente, Voggenberger, Bernhard, Berghammer, Wilhelm, Klotz, Daniel, Böck, Sebastian, Klambauer, Günter & Hochreiter, Sepp 2026 TiRex-2: generalizing TiRex to multivariate data and streaming. *arXiv preprint arXiv:2607.01204* .

Pope, Stephen B 2001 Turbulent flows. *Measurement Science and Technology* **12** (11), 2020–2021.

Sauer, Tim, Yorke, James A. & Casdagli, Martin 1991 Embedology. *Journal of Statistical Physics* **65** (3–4), 579–616.

Skinner, J. W., Lawrence, A. & Callies, J. 2025 Characterizing ocean flows with the scattering transform. *arXiv preprint* ArXiv:2505.00819.

Smith, Luke, Fukami, Kai, Sedky, Girguis, Jones, Anya & Taira, Kunihiko 2024 A cyclic perspective on transient gust encounters through the lens of persistent homology. *J. Fluid Mech.* **980**, A18.

Solera-Rico, Alberto, García-Caspueñas, Patricia, Sanmiguel Vila, Carlos & Discetti, Stefano 2025 The balance between compactness and forecast accuracy of data-driven latent-space reduced-order models in controlled wake flows. Under review, Journal of Fluid Mechanics.

Solera-Rico, Alberto, Sanmiguel Vila, Carlos, Gómez-López, M., Wang, Y., Almashjary, A., Dawson, S. T. M. & Vinuesa, R. 2024 β-variational autoencoders and transformers for reduced-order modelling of fluid flows. *Nat. Commun.* **15**, 1361.

Taira, Kunihiko 2026 Extreme aerodynamics: a data-driven perspective. *Physical Review Fluids* DOI 10.1103/8pr4-txmt; arXiv:2511.12889.

Takens, Floris 1981 Detecting strange attractors in turbulence. In *Dynamical Systems and Turbulence, Warwick*

1980 (ed. David Rand & Lai-Sang Young), *Lecture Notes in Mathematics*, vol. 898, pp. 366–381. Berlin: Springer.

Taylor, G. I. 1918 On the dissipation of eddies. *Reports and Memoranda, Advisory Committee for Aeronautics* (598).

Wang, Zhou, Bovik, Alan C., Sheikh, Hamid R. & Simoncelli, Eero P. 2004 Image quality assessment: from error visibility to structural similarity. *IEEE Transactions on Image Processing* **13** (4), 600–612.

Zhong, Yonghong, Fukami, Kai, An, Byungjin & Taira, Kunihiko 2023 Sparse sensor reconstruction of vortex-impinged airfoil wake with machine learning. *Theoretical and Computational Fluid Dynamics* **37** (2), 269–287.